

UNIVERSITI TEKNOLOGI MARA

**MYSAWIT: PALM FRUIT RIPENESS
DETECTION WITH TRACEABILITY
SYSTEM**

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**BACHELOR OF COMPUTER
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Thesis submitted in fulfillment
of the requirements for the degree of
Bachelor of Computer Science (Hons.)

**College of Computing, Informatics and
Mathematics**

August 2024

AUTHOR'S DECLARATION

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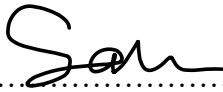
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ABSTRACT

MySawit is a mobile application that allows you to check the ripeness of a palm fruit just from your phone. The application can also grade the palm fruit by comparing the weight of the palm fruit and the ripeness stage it is currently at. This allows for a more accurate grading and will allow only good quality of oil to be produced. In this project, we identified the problem statements and objectives of what this application will be. Aside from that, we have made literature review where we made analysis of the overview of palm fruit and identifying the ways on detecting the ripeness of palm fruit in different stages. Comparing the algorithms and similar applications is crucial in identifying what are the best methods to use and upgrades that we can do for MySawit. We also have a look at the methodology that is suitable for this project which is the waterfall model. We also explore on how the project will be executed by doing the system design and implementations of this project. The result and discussion are shown to give insight into the development of this mobile application. Lastly, the conclusion and recommendations show what the outcome of the project is and the strengths and limitations of the develop mobile application. The recommendation explains on what improvements can be made on this project in the future.

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TABLE OF CONTENTS

	Page
AUTHOR’S DECLARATION	ii
ABSTRACT	iii
ACKNOWLEDGEMENT	iv
TABLE OF CONTENTS	v
LIST OF TABLES	ix
LIST OF FIGURES	x
LIST OF ABBREVIATIONS	xii
CHAPTER ONE INTRODUCTION	1
1.1 Background of Study	1
1.2 Problem Statement	1
1.3 Objectives	2
1.4 Significance of Study	3
1.5 Significance of Study	3
1.6 Summary	4
CHAPTER TWO LITERATURE REVIEW	6
2.1 Palm Fruit Overview	6
2.1.1 Types of Land for Palm Fruit Farming	6
2.1.2 Climate	6
2.1.3 Location	7
2.2 Ripeness Identification	7
2.2.1 Method to Harvest Palm Fruit	8
2.2.2 Quality of Palm Fruit Depends on Ripeness	8
2.2.3 Identifying the Ripeness	9
2.3 Machine Learning Algorithm in Identification of Palm Fruit	11
2.3.1 Convolution Neural Network (CNN)	12
2.3.2 Support Vector Machine (SVM)	12
2.3.3 K – Nearest Neighbour (KNN)	12

2.3.4	Comparison Between the Algorithms	13
2.3.5	Summary	14
2.4	Traceability System	14
2.4.1	What is Traceability to Plantation (TTP)	15
2.4.2	Why is TTP important for Palm Fruit	15
2.4.3	Quick Response (QR) Code Technology in TTP	16
2.5	Similar Applications	17
2.5.1	Google Lens	17
2.5.2	9 Angle Crop	20
2.5.3	PaLife	21
2.5.4	Comparison Between Applications	22
2.6	Summary	23
CHAPTER THREE RESEARCH METHODOLOGY		24
3.1	Waterfall Model	24
3.2	Requirement Gathering and Analysis	26
3.2.1	System User Identification	26
3.2.2	User Requirements	27
3.3	System Design	28
3.3.1	System Architecture	28
3.3.2	System Flowchart	29
3.3.3	Algorithm Design	31
3.3.4	Entity Relationship Diagram (ERD)	33
3.3.5	Use Case Diagram	34
3.3.6	Use Case Description	35
3.3.7	User Interface	38
3.4	System Implementation	45
3.4.1	Hardware Requirements	45
3.4.2	Software Requirements	46
3.5	Testing	47
3.5.1	Test Case	47
3.6	Gantt Chart	51
3.7	Summary	52

CHAPTER FOUR RESULTS AND DISCUSSION	53
4.1 SQLite Database	53
4.1.1 Database and Table Creation	53
4.1.2 Data Insertion and Retrieval	54
4.1.3 Generating QR Code	55
4.2 Ripeness Detection Implementation	56
4.3 Application Interface	57
4.3.1 Log in Interface	57
4.3.2 Register Interface	58
4.3.3 Check Ripeness	60
4.3.4 Add Palm Fruit Batch Information	61
4.3.5 Palm Fruit Batch Information	63
4.3.6 Quality Grading Interface	64
4.3.7 Grading Information Interface	65
4.4 Functionality and Results	66
4.4.1 User Login Testing	66
4.4.2 Palm Fruit Ripeness Detection Testing	68
4.4.3 Insert Batch Information Testing	69
4.4.4 Displaying Palm Fruit Batch Information Testing	69
4.4.5 Quality Grading of the Palm Fruit Batch Testing	70
4.5 Discussions	71
4.6 Summary	72
 CHAPTER FIVE CONCLUSION AND RECOMMENDATION	 73
5.1 Revisiting Objectives	73
5.1.1 Development of Machine Learning Algorithm to Detect the Ripeness of Palm Fruit	73
5.1.2 Implementation of traceability system function on the transportation of Palm Fruit	74
5.1.3 Testing the Functionality of the Mobile Application	74
5.2 Strengths and Limitations	75
5.2.1 Strengths	75
5.2.2 Limitations	75
5.3 Future Work Recommendations	75

5.4	Summary	76
	REFERENCES	77
	APPENDICES	82

LIST OF TABLES

Tables	Title	Page
Table 2.1	Comparison Between the Algorithms	13
Table 2.2	Comparison Between Similar Applications	22
Table 3.1	Activities and Outcome of Waterfall Model	25
Table 3.2	Login Use Case Description	35
Table 3.3	Check Palm Fruit Ripeness Use Case Description	36
Table 3.4	Enter Palm Fruit Information Use Case Description	36
Table 3.5	Display Palm Fruit Information Use Case Description	37
Table 3.6	Edit Palm Fruit Information Use Case Description	37
Table 3.7	Hardware Requirements	45
Table 3.8	Software Requirements	46
Table 3.9	Programming Language Used	46
Table 3.10	Login Test Case	48
Table 3.11	Palm Fruit Ripeness Detection Test Case	48
Table 3.12	Enter Palm Fruit Information Test Case	49
Table 3.13	Display Palm Fruit Information and Generate Report Test Case	49
Table 3.14	Grade Palm Fruit Quality Test Case	50
Table 4.1	User Login Functionality	67
Table 4.2	Palm Fruit Ripeness Detection Functionality	68
Table 4.3	Insert Palm Fruit Information Functionality	69
Table 4.4	Display Palm Fruit Information	70
Table 4.5	Determine Quality of Palm Fruit	71

LIST OF FIGURES

Figures	Title	Page
Figure 2.1	Under Ripe Palm Fruit	9
Figure 2.2	Unripe Palm Fruit	10
Figure 2.3	Ripe Palm Fruit	10
Figure 2.4	Over Ripe Palm Fruit	11
Figure 2.5	Example of TTP	15
Figure 2.6	Search Function in Google Lens	17
Figure 2.7	Translate function of Google Lens	18
Figure 2.8	Scanning Text in Real Time	19
Figure 2.9	Interface for 9 Angle Crop for palm fruit detection	20
Figure 2.10	Interface of PaLife	21
Figure 3.1	Waterfall Model Structure	24
Figure 3.2	System Architecture of the project	28
Figure 3.3	System Flowchart	29
Figure 3.4	System Flowchart (Continued)	30
Figure 3.5	Entity Relationship Diagram	33
Figure 3.6	Use Case Diagram	34
Figure 3.7	Login Interface	38
Figure 3.8	Farmer Main Menu Interface	39
Figure 3.9	Ripeness Detection Interface	40
Figure 3.10	Enter Palm Fruit Information Interface	41
Figure 3.11	Staff Main Menu Interface	42
Figure 3.12	Quality Grading Interface	43
Figure 3.13	Palm Fruit Information Interface	44
Figure 4.1	Coding Segment to Create Database in SQLite	53
Figure 4.2	Coding Segment to Create Tables in SQLite	54
Figure 4.3	Coding Segment to Insert the Palm Fruit Batch Information	54
Figure 4.4	Coding Segment to Insert Palm Fruit Batch Information in HelperClass	55
Figure 4.5	Coding Segment to Generate QR Code	55

Figure 4.6	Coding Segment for Ripeness Detection Implementation	56
Figure 4.7	Login Interface	57
Figure 4.8	Error Message of Incorrect Credentials	58
Figure 4.9	Farmer Registration	58
Figure 4.10	Staff Registration	59
Figure 4.11	Check Ripeness Interface	60
Figure 4.12	Ripeness Successfully Detected	60
Figure 4.13	Add Batch Information Interface	61
Figure 4.14	Dropdown Menu with Location	62
Figure 4.15	Palm Fruit Batch Information Interface	63
Figure 4.16	Quality Grading Interface	64
Figure 4.17	Determine the Quality of Palm Fruit	64
Figure 4.18	Grading Information Interface	65

LIST OF ABBREVIATIONS

Abbreviations

ML	Machine Learning
AI	Artificial Intelligence
SVM	Support Vector Machine
CNN	Convolution Neural Network
KNN	K-Nearest Neighbour
TTP	Traceability to Plantation
GPU	Graphics Processing Unit
IDE	Integrated Development Environment

CHAPTER ONE

INTRODUCTION

This chapter provides the background and rationale for the study. It gives details of palm fruit ripeness detection with traceability system followed by highlighting the problem statement that led to the study. Then, this chapter continues with the aim and objectives, before stating the scope and significance of the study. Lastly, it ends with a summary of this chapter.

1.1 Background of Study

The palm oil industry is one of the biggest industries in Malaysia due to the production of its oil that have many uses in daily live products like food, cosmetics and soaps (Septiarini, A. et al., 2019). The Malaysian palm oil mill ranks as the fourth major contributor to the country's economy and constitutes a significant portion of Malaysia's Gross National Income. (T. S. Hong et al., 2021). In the palm oil industry, the quality and production of oil extraction highly depends on the ripeness of the palm fruit. This is because when extracting the oil when the palm fruit is ripe will ensure high oil extraction rate (Ghazali, S. A. et al., 2019). Aside from the ripeness of the palm fruit, the harvesting phase is also important due to the factors that could happen when the fruits are being transferred in batches from the farm to the factory. As of now there has been a lack of visibility to track which batches of palm fruit could produce good quality palm oil (Hirbli, T., 2018). Hence, image processing using machine learning (ML) will be implemented to effectively identify the ripeness of palm fruit (Septiarini, A. et al., 2019) and use QR code to allow the staff to check the information of the palm fruit oil.

1.2 Problem Statement

As we all know, the ripeness of the palm fruit is key to ensure a good quality palm oil. As of now, the method used by farmers to harvest these palm fruits are by using the manual method where they observe and sort the palm fruit base on the colour of the fruit skin layer (T. S. Hong et al., 2021).

According to Ghazali, S. A. et al. (2019), mistakes in manual visual inspection might happen during harvesting because when inspecting the ripeness of the palm fruit, different farmers may interpret the ripeness differently which will produce inconsistent outcome of the palm fruit harvesting. Other than that, this problem is also considered as time consuming and less efficient due to the difficulty that arises during harvesting for example the palm tree being too tall to be reached by normal human height and is hard to determine the ripeness of the fruit (Suharjito, 2021).

Another problem that arises is during the transportation of palm fruit from the farm to the oil factory poses a significant challenge to maintaining the quality of the fruit, as it is prone to get degraded. This is due to the fruits susceptibility to damage like crushing or bruising to the fruit during transportation (Vikaliana et al., 2021). Poor handling practices of the palm fruit can cause harm to the fruit's condition and the quality of the oil to degrade. Additionally, environmental conditions further exacerbate the risk of quality deterioration. Heavy rain can be quite bad during transportation. Excessive moisture during transportation not only increases the likelihood of damage to the fruit but also creates an environment suitable to the growth of microorganisms, potentially leading to spoilage and a decline in the overall quality of the palm fruit (Hirbli, T., 2018).

1.3 Objectives

In order to overcome the problem to ensure quality of the palm fruit, this project aims to develop a palm fruit ripeness detection with traceability system specifically for mobile application to do easily detect the ripeness of the palm fruit. Thus, the objectives include to:

- a. To develop a suitable machine learning algorithm to detect the ripeness of the palm fruit.
- b. To implement a traceability system to track the process of a palm fruit.
- c. To test the functionality of palm fruit ripeness detection with traceability system.

1.4 Significance of Study

i. Area

The project will focus on palm fruit factory in Sabah, Malaysia. The selection of this geographic focus comes from the significant impact of the palm fruit oil business in the area, as well as the distinct advantages and obstacles linked to palm oil production in Sabah.

ii. Target User

- **Palm fruit farmers**

Help the farmers at the palm fruit farm to identify the ripeness of the palm fruit during harvesting by using their mobile phones and display to the farmers whether the fruits are suitable to harvest or not.

- **Factory management staff**

Managing the batches of palm fruit by classifying ripe and unripe fruits and updating the system of the oil production.

iii. Method of Use

This program will be built using image processing with the help of machine learning to identify the ripeness of the palm fruit and use QR code to allow staff to grade the palm fruit oil batches that arrive to the mill. Next, the platform of this program will be built on a mobile application as it can be convenient for the users to use. The mobile application will implement the image processing to help users identify the ripeness of the palm fruits.

1.5 Significance of Study

The suggested solution focuses on improving the quality of palm oil by assisting farmers in accurately determining the ripeness of palm fruit. This is crucial because the ripeness of the fruit directly affects the quality of the oil produced. By providing valuable support

to farmers in this aspect, the solution aims to enhance the overall output of high-quality palm oil.

Furthermore, the system plays a key role in helping workers monitor and ensuring compliance to established standard of procedure during the palm oil production process. This means ensuring that each step in the production follows the recommended procedures, contributing to a consistent and reliable quality output.

Additionally, the solution aids factory workers in a systematic approach to trace and assess the quality of palm fruits on a batch-by-batch basis. This involves a detailed examination of each batch, allowing for a comprehensive system for quality control and assurance throughout the entire production cycle. This meticulous approach is essential to identify and address any potential issues, ultimately ensuring the production of consistently high-quality palm oil.

1.6 Summary

This chapter describes the importance of why the palm oil industry is a major contributor to Malaysia's economy, with various uses in daily products. Quality and oil production depend on the ripeness of the palm fruit. Furthermore, there's a need to track batches of palm fruit for quality control. To address this, image processing with machine learning will identify the fruit ripeness and traceability to track the information of the palm fruit to allow the staff to grade the palm fruit oil to determine whether it is good for oil production and what other uses it is for.

The ripeness of palm fruit is crucial for high-quality palm oil production, but manual harvesting methods lead to inconsistencies among farmers which is time-consuming and inefficient. Transportation challenges also risk fruit quality degradation. To address these issues, a system that is built on a mobile application platform for palm fruit ripeness detection and traceability system will be developed. The focus of this project will be at Sabah, Malaysia due to its geographic focus which comes from the significant impact of the palm oil business in the area. Objectives include implementing a machine learning algorithm, traceability system, and evaluating functionality. The system benefits farmers during harvesting, allowing mobile detection of fruit ripeness, and assists factory workers in managing batches. The proposed solution aims to enhance palm oil quality by ensuring accurate ripeness detection, promoting compliance with production standards, and establishing a comprehensive quality control framework

throughout the production process.

CHAPTER TWO

LITERATURE REVIEW

2.1 Palm Fruit Overview

This section of the report provides a general overview of the type of land where palm fruits are usually planted, suitable climate to cultivate palm fruits and the location of where these palm fruits are usually located.

2.1.1 Types of Land for Palm Fruit Farming

Knowing the type of land that can plant palm fruit is crucial in making sure it grows in good condition. Palm fruits are plants that can adapt to a wide range of tropical soil like deep peat and sand substrates (Meijaard et al., 2020). The soil needs to have the right amount of moisture for it to thrive, which means that frequent rainfall and enough water supply is important to keep the palm fruits in good condition (Abubakar et al., 2022).

Planting the palm fruits needs to be in a big area with open space for the plant to survive (Abubakar et al., 2022). This is due to the plant could grow very big and reaching up to very tall heights, it needs enough sunlight for photosynthesis making an open space to be suitable to plant the palm fruit. This shows that the plant has a versatile nature that allows it to grow in different soil compositions that are commonly found in tropical regions and need the right amount of water and an open space to grow well.

2.1.2 Climate

To farm palm fruits, it is good to know the kind of climate that is suitable where the palm fruit can thrive and produce the finest quality of fruits. Palm fruits need high humidity levels as it benefits the production of the fruits which makes humid tropics to be an ideal condition for palm fruit growth (Meijaard et al., 2020). High humidity levels reduce water loss through transpiration and can benefit the plant's overall health. It may also have an effect on the efficiency of gas exchange in the leaves, allowing for optimal photosynthetic rates.

Furthermore, palm fruits thrive in warm climates where the temperature to

cultivate the plant is between 18 and 38 °C (Shevade & Loboda, 2019). Warm temperatures aid the plant to increase metabolic activities and development. High solar radiation helps the plant to get enough sunlight that is required for photosynthesis process that turns sunlight into energy for the palm fruit that can help for a healthy and productive palm fruit (Meijaard et al., 2020).

2.1.3 Location

The location of palm fruit cultivation is important due to its specific environmental requirements. The plant is commonly found at southeast Asia where Malaysia and Indonesia being the largest countries to produce and export oil palm fruit to other countries like India, China, and the United States (Abubakar et al., 2022).

Malaysia is suitable for palm fruit cultivation because of its tropical climate that always have high temperatures throughout the year. High temperatures and humidity combine to provide a tropical climate that supports diverse flora and varied ecosystems (Abubakar et al., 2022). Furthermore, Malaysia receives abundant of rainfall throughout the year especially during monsoon season that happen during May to September and November to March (Abubakar et al., 2022).

2.2 Ripeness Identification

Palm fruits have organic properties like carotenoid and lycopene which are used in forming the colour of the plant (Hong et al., 2021). These organic compounds not only improve the visual appearance of the palm fruit but also affect the quality and nutritional content of the palm fruit. Recognizing the appropriate level of ripeness of palm fruit is crucial since only ripe palm fruit will provide the maximum amount of oil palm (Ibrahim et al., 2018). This section reveals the method to harvest the palm fruit, explain how quality of palm fruit depends on the ripeness and the identification of ripe palm fruits.

2.2.1 Method to Harvest Palm Fruit

The first step in optimizing oil extraction is selecting the right palm fruit to harvest and may only be collected on palm fruit estates when the palm trees achieve maturity three years after planting (Lai et al., 2023). The tree maturity is important since it directly impacts the oil content and quality of the harvested fruit. Younger trees may not produce enough oil, and the oil they do produce may not match the acceptable quality criteria.

Currently, the ripeness of the palm fruit is solely determined by manual visual examination, which is done by palm fruit industry workers, also known as human graders, who use their eyesight to identify the ripeness (Herman. 2019). During the harvesting phase of 10-14 days, field workers will look for palm fruit trees with an abundance of detached fruitlets on the ground underneath and the workers will then visually identify ripe fruits based on the color of the fruit as well as the sockets made when the fruitlets fall to the ground. (Lai et al., 2023). Fruitlet detachment is a natural process that occurs when the palm fruits are mature. The harvested palm fruits will then be gathered and sent to a palm fruit mill to extract the oil. To assure the quality of the fruit, they are generally sent to the mill within 24 hours of harvesting (Lai et al., 2023).

2.2.2 Quality of Palm Fruit Depends on Ripeness

Quality plays an important role for the oil palm industry since it is thought that quality is the main factor in achieving success in today's competitive market (Hong et al., 2021). A study was conducted to investigate the association between oil concentration and palm fruit quality. According to this study, good grade palm fruit has the most oil content, whereas low quality ones have the least (Ghazali, 2019). Monitoring and managing the ripeness is important to ensure that the harvesting takes place at the optimum stage which maximizes both the quantity and quality of the produced oil. Palm fruit that has low quantities of free fatty acids is said to be of good grade. Free fatty acids can have an adverse influence on the oil's taste, fragrance, and stability (Ghazali, 2019).

2.2.3 Identifying the Ripeness

The appearance of the palm fruit is a very crucial aspect determining the ripeness in the palm fruit industry (Ghazali, 2019). The visual examination of palm fruit provides crucial indicators that determine the level of ripeness and is able to be harvested (Ghazali, 2019). Identifying the palm fruit ripeness can be categorized in four different classes which are unripe, under ripe, ripe, and over ripe (Sulaiman et al., 2020). This categorization will be based on the colour of the palm fruit skin and other characteristics of the palm fruit such as number of sockets on the palm fruit (Lai et al., 2023). Mentioning the quality is also significant to know which stages of palm fruit will produce good quality oil production (Sulaiman et al., 2020).

2.2.3.1 Under Ripe Palm Fruit



Figure 2.1 Under Ripe Palm Fruit

(Source: MPJ Palm Oil)

Under ripe palm fruits have an orange and reddish colour which takes up most of the surface (Sulaiman et al., 2020). The palm fruit have less than ten loose sockets which is a sign that the palm fruit are still young and not suitable to harvest for oil production (Ibrahim et al., 2018). Furthermore, underripe palm fruits are known to have lower oil quality compared to unripe ones, making them less desirable for oil extraction (Ibrahim et al., 2018).

2.2.3.2 Unripe Palm Fruit



Figure 2.2 Unripe Palm Fruit

(Source: MJM Palm Oil)

Unripe palm fruit have a black and dark purple surface on the overall surface of the fruit but have a little bit of yellowish colour in the inner part of the fruit (Ibrahim et al., 2018). Unripe palm fruits have less oil than fully ripe ones which results in them being less ideal for oil extraction since the amount of oil produced is not optimal (Ghazali, 2019).

2.2.3.3 Ripe Palm Fruit



Figure 2.3 Ripe Palm Fruit

(Source: MJM Palm Oil)

Ripe palm fruits have an orange with reddish hue on the overall surface (Ibrahim et al., 2018). They contain at least ten loose sockets and have most of the fruits intact to the

palm fruit (Hong et al., 2021). Ripe palm fruits are usually sent to the mill within 24 hours after getting harvested (Lai et al., 2023). Due to their high oil content, ripe palm fruits are ideal for oil production.

2.2.3.4 Over Ripe Palm Fruit



Figure 2.4 Over Ripe Palm Fruit

(Source: MJM Palm Oil)

An over ripe palm fruit have a dark red surface colour and have most of the fruits detached from the palm fruit (Ibrahim et al., 2018). Over ripe fruits may have less amount of oil than fruits taken at their peak ripeness (Ghazali, 2019). This can lead to decreased oil extraction efficiency.

2.3 Machine Learning Algorithm in Identification of Palm Fruit

Machine Learning (ML) is an area of artificial intelligence (AI) that focuses on the development of algorithms and models that allow computers to learn from and make predictions or judgments based on the data (Bi et al., 2019). The basic idea underlying machine learning is to enable computers to improve their performance on a certain activity over time without being explicitly taught (Liu, 2020). This section shows the ML algorithm that are used for image recognition and comparison between them.

2.3.1 Convolution Neural Network (CNN)

A convolutional neural network is a deep learning technique that takes an input image and assigns meaning to various objects in the image, allowing them to be separated from one another (Bhatt et al., 2021). The CNN algorithm is especially efficient in computer vision applications that require the recognition of patterns, features, or objects in visual data (Alzubaidi et al., 2021). Model architecture of CNN is crucial for improving the performance of various applications and is affected by visual processing in the human brain. (Alzubaidi et al., 2021).

2.3.2 Support Vector Machine (SVM)

Support Vector Machine (SVM) uses mathematical computer model that is utilized for problem classification, mainly in the field of classification and regression (Chandra & Bedi, 2018). It seeks to discover a hyperplane in feature space that optimally separates data points from distinct classes (Liu et al., 2020). The method operates by transferring input data points into a higher-dimensional space and searching for the best hyperplane that separates the classes the most (Liu et al., 2020).

2.3.3 K – Nearest Neighbour (KNN)

The K-Nearest Neighbours (KNN) algorithm is a supervised ML algorithm that is derived from the geometric measurement that may be used for regression and classification tasks that are accomplished by computing the distance between distinct feature values in the model (Taunk et al., 2019). It is a straightforward and user-friendly approach that predicts the majority class or average value of the k-nearest data points in the feature space (Liu et al., 2020).

2.3.4 Comparison Between the Algorithms

Criteria	CNN	SVM	KNN
Application	Image recognition, video analysis	Classification and regression tasks	Classification and regression tasks
Accuracy	Highest for image and video analysis	Moderate but can be high depends on the kernel and parameters	Moderate due to sensitive to noise and outliers
Principle	Simulates local perception of a multi – layer to achieve input – to – output mapping.	VC dimension theory for non-linear classification problems and structural risk minimization theory for high dimensional feature space classification problems	Geometric measurement for regression and classification
Amount of dataset	Requires large datasets	Requires moderate datasets	Requires smaller datasets
Training time	Longer training time	Depends on the complexity of the problem	Quite Quick

Table 2.1 Comparison Between the Algorithms

(Adapted: Liu et al., 2020)

Based on the table above, CNN is shown to be used for image processing like image recognition and even video analysis which have the highest accuracy rate compared to SVM and KNN.

Aside from that SVM and KNN have the same use which is classification and regression tasks. Both have different accuracy rate where SVM is shown to have moderate to high accuracy depending on the kernel and parameters. KNN however is moderate due to sensitive to noise and parameters.

2.3.5 Summary

Based on the comparison of all three algorithms, the CNN algorithm is found to be the most suitable for palm fruit ripeness identification due to its excellent use in image related tasks and have a convolution layer that are effective in capturing. CNN also have the highest accuracy rate out of the SVM and KNN.

2.4 Traceability System

What is traceability system? Well traceability system is a system that has the capability to track and trace information which can create a sense of transparency in supply chain (Sunny, 2020). Nowadays, there has seem to be an increased number of organizations that have supported research on the implementation of traceability system which shows how important it is to have traceability system (Kroll, 2021). This section will give the explanation of traceability to plantation (TTP), the importance of TTP and the technology used for traceability which is Blockchain technology.

2.4.1 What is Traceability to Plantation (TTP)



Figure 2.5 Example of TTP

(Source: The Sustainable Palm Oil Choice, 2023)

According to Tolls For Transformation, an online website that specialise in traceability, explains that Traceability to Plantation or TTP for short is the capability to keep up with the palm fruits from the production stage at the palm fruit farm to the plantation mill to process the fruits (*Earthworm Foundation Tool for Mills Guide to Achieving Traceability to Plantation, n.d.*). Traceability of agricultural goods is designed to offer information on all or part of the agricultural product supply chain, from harvesting to transportation, storage, distribution, and sales, to ensure the quality and safety of products disseminated or eaten by consumers (Vikaliana et al., 2021).

2.4.2 Why is TTP important for Palm Fruit

Some researcher has said that consumers concerned about the quality of their food product is very important and have increased in the recent years (Mirabelli & Solina, 2020). This is due to some concerns that arise as there is a link between diet and health which makes the consumers want to maintain a healthy lifestyle with high quality and nutritional foods (Vikaliani et al., 2021).

2.4.3 Quick Response (QR) Code Technology in TTP

Ever since the release of mobile smartphones, QR codes has become a famous 2D barcodes that has been used by a lot of people as of recent years (Wahsheh & Luccio, 2020). It offers a lot of functionality allowing people to be able to open websites, watch videos or even listen to music with just a scan of the QR code in their own smartphones (Focardi et. al., 2019). With the use of this technology, the implementation of it to provide traceability to its users are wide and simple (Wahsheh & Luccio, 2020). The way it can be used is by ensuring that each of the palm fruit batches has its own QR code that contains details of the fruits like, the harvesting date of the palm fruit, where it originated and the quality of the fruits when it was harvested and after the palm fruit oil is produced (Gao et. al., 2019).

2.5 Similar Applications

2.5.1 Google Lens

Google Lens is an application that was invented by Google. It was developed using AI to perform many tasks. The main purpose of this application is to help users to easily take a picture and get the information related to the picture. Here are some examples of what Google Lens could do:

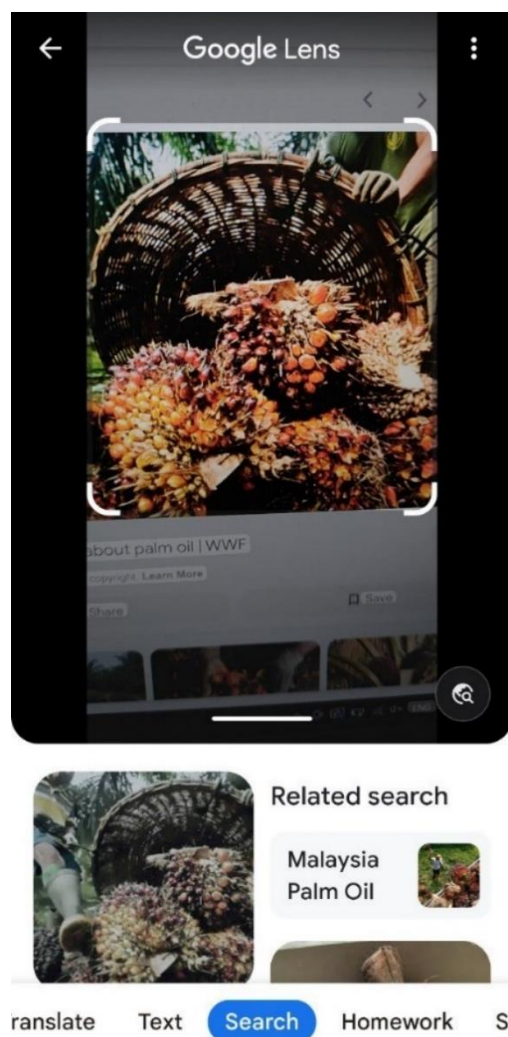
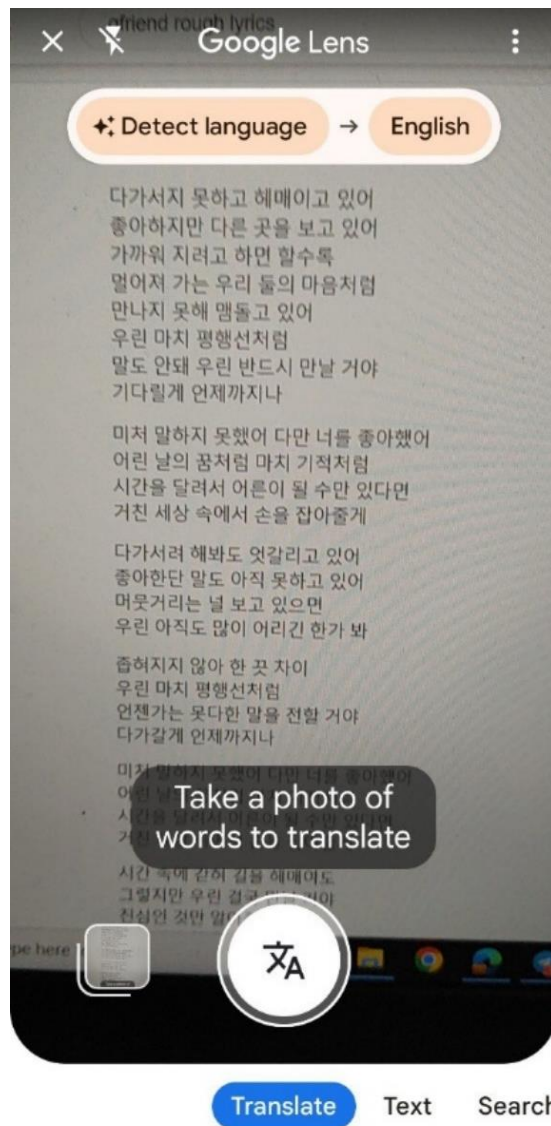


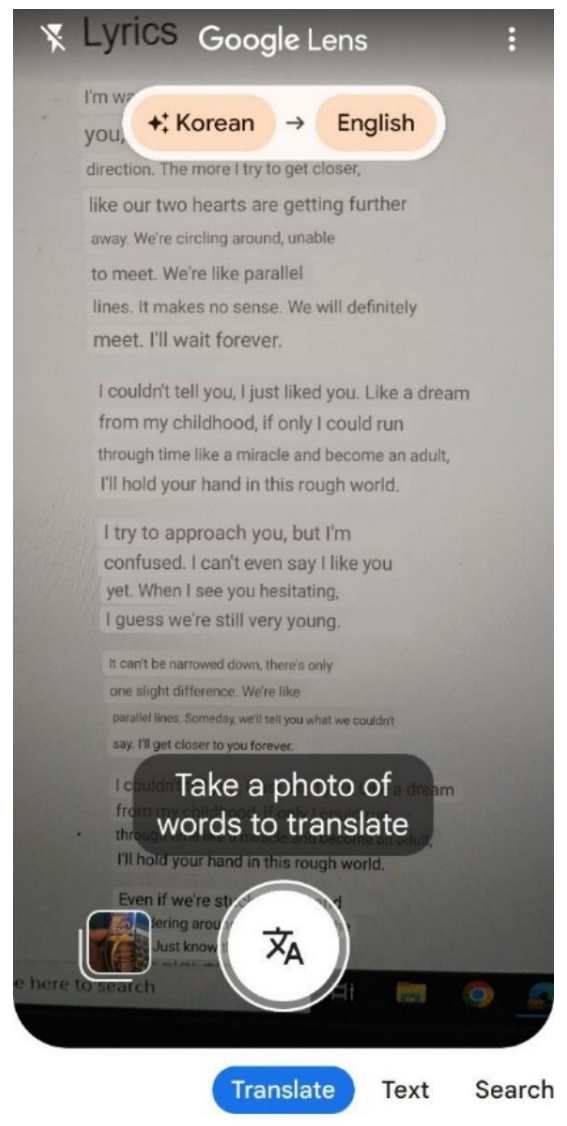
Figure 2.6 Search Function in Google Lens

Search function based on the picture taken on the app or found in the gallery app.

- Translate other languages to another language in real time or picture taken from gallery.



(a) Before translation



(b) After translation

Figure 2.7 Translate function of Google Lens

(Source: Google Play Store, 2023)

- Scan text in real time or picture taken from the gallery.

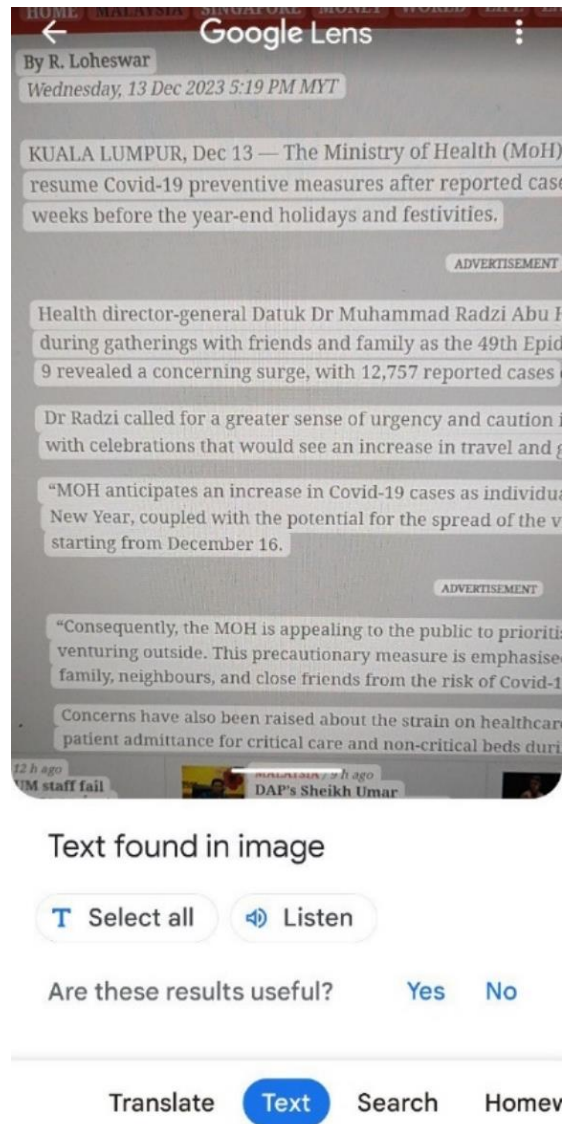


Figure 2.8 Scanning Text in Real Time

(Source: Google Play Store, 2023)

The application is a generally good to use for searching things, translate foreign text to other languages, and scan text which is done by using a single picture or in real time. Aside from that, the application is not suitable for identifying the ripeness of fruits as it lacks the function to allow the users to have a specific instruction for it to search.

2.5.2 9 Angle Crop

Angle Crop is an application that was created by Suharjito et al. (2021). It was developed with the aim to snap a picture of a palm fruit and identify the ripeness base on three categories which are unripe, ripe and over ripe. Here are some of the features for 9 Angle Crop:

- Can classify the ripeness of palm fruit which are unripe, under ripe, ripe, over ripe, abnormal and empty palm fruit (Suharjito et al., 2021).

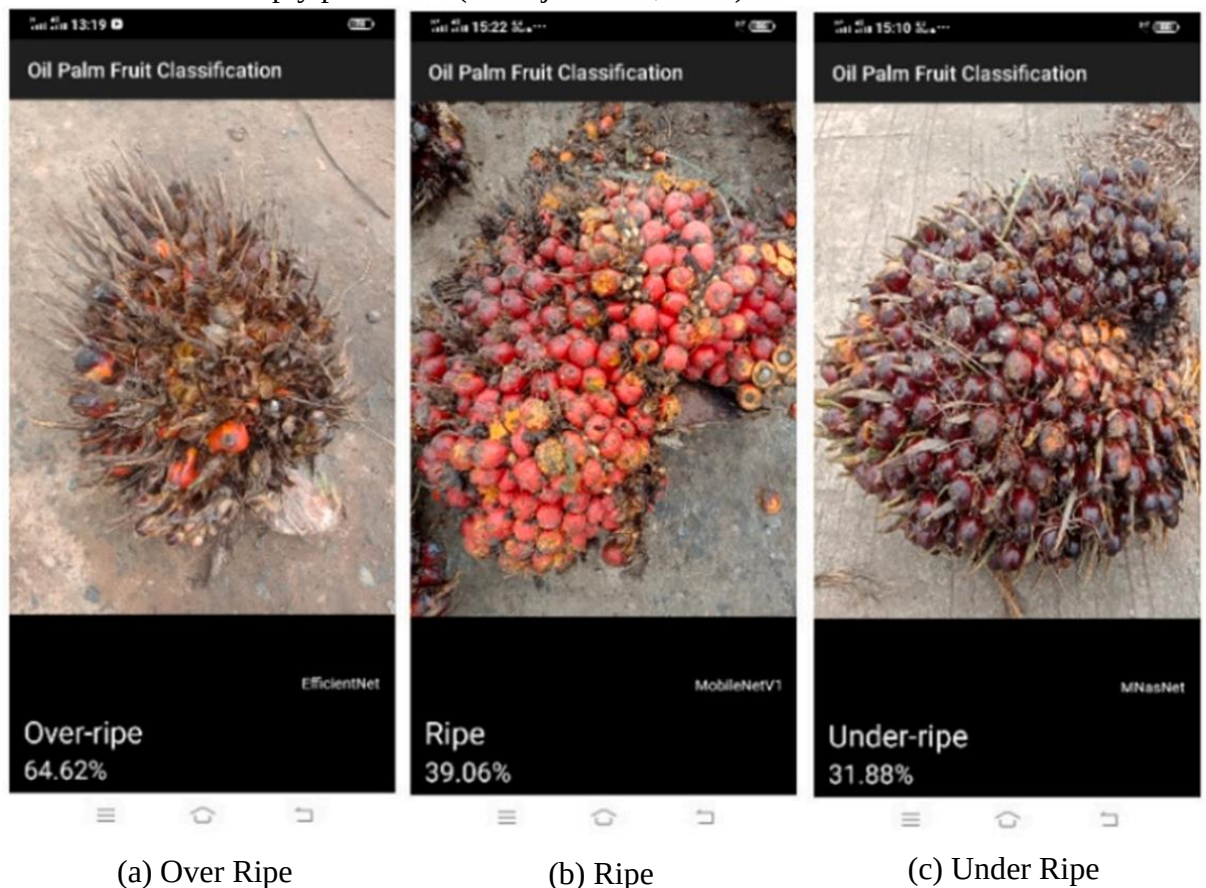


Figure 2.9 Interface for 9 Angle Crop for palm fruit detection
(Source: Oil palm fresh fruit bunch ripeness classification on mobile devices using deep learning approaches)

- Uses the deep learning models such as MobileNetV1, MobileNetV2, NASNet Mobile and EfficientNetBO to detect the ripeness of palm fruits.
- Combining transfer learning and 9 angle crop method prove to increase accuracy of MobileNetV1.

2.5.3 PaLife

PaLife is a mobile application that was created by Echalar and Subinon (2018). The main purpose of this application is to provide detection of rice health condition by using image processing and analysis of pigment of the rice (Echalar & Subinon, 2018). Here are some of the features that the application provides:

- Uses CNN and colour segmentation algorithms to identify the condition of the rice.
- Effective detection for problems.

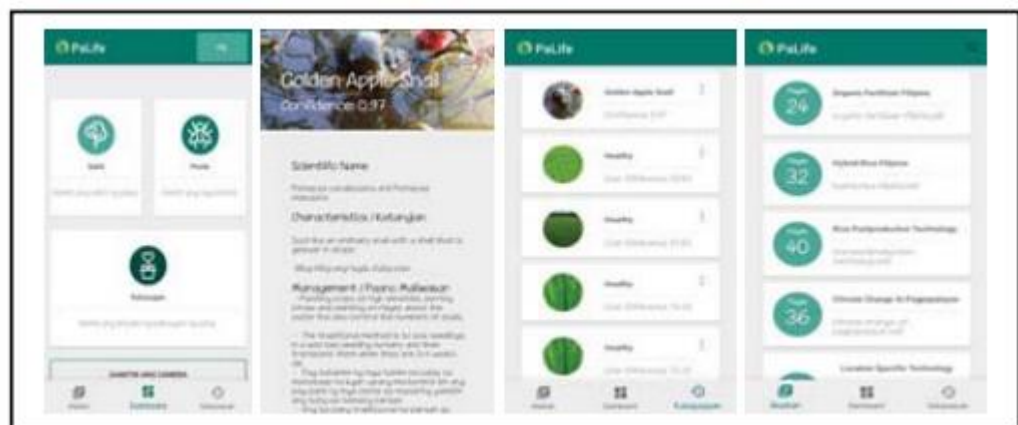


Figure 2.10 Interface of PaLife

(Source: PaLife: A Mobile Application for Palay (Rice) Health Condition Classification utilizing Image Processing and Pigment Analysis towards Sustainability of Palay Production)

2.5.4 Comparison Between Applications

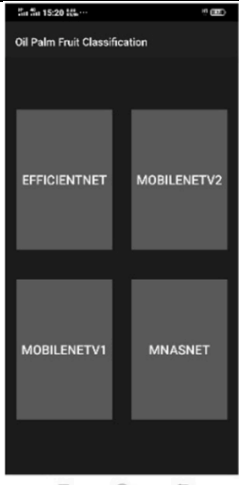
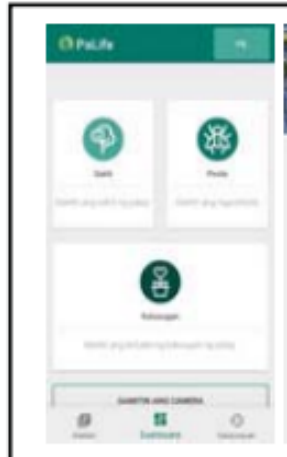
Criteria	Google Lens	9 Angle Crop	PaLife	MySawit
Interface				No interface
ML algorithm used	CNN	CNN	CNN, Local Binary Pattern Histogram and Color Segmentation algorithms	CNN
Function	To scan, copy text image and translate text	Detecting the ripeness of palm fruit	Detection of rice health condition	Detect palm fruit ripeness base on color and provide traceability to plantation
Operating System type	Android	Android	Android	Android
Traceability	No	No	No	Yes
Qr Code	Yes	No	No	Yes

Table 2.2 Comparison Between Similar Applications

Based on the table above, all the applications compared has a very similar ML algorithm which is using CNN for the image processing aspect of the applications. This is due to the high recognition rate that CNN provides which helps to identify the images very accurately.

Other than that, all the applications have very similar functions which is to use image recognition to identify the objects. Only Google Lens does not have a specific function like other application which is used to detect the ripeness or health condition.

For the OS type, all the applications use android as the base operating system. Furthermore, in terms of traceability only MySawit, an upcoming application, will have traceability function to trace the whereabouts of the palm fruits and categorize good and bad palm fruit quality for production. MySawit is also the only application that applies QR code technology which a technology commonly used for traceability systems.

2.6 Summary

In chapter 2, we have seen the literature review on the overview of palm fruit which explains the details about the type of land that is suitable for palm fruit farming, the climate suitable to cultivate palm fruits and the location where palm fruits are commonly founded. We also explore in the identification for palm fruit ripeness where it explains the methods used to harvest palm fruit, relate how quality is determined by the ripeness of the palm fruit, and ways to identify the ripeness of the palm fruit based on colour.

Furthermore, the literature review also analyzes the machine learning algorithm used for ripeness identification of the palm fruit where we explore the three main algorithms which are CNN, SVM and KNN. Comparison between these three algorithms is also shown to see which one the suitable algorithm is to use that can produce good accuracy.

Next, we also explore about traceability system where explanation of what is TTP, the importance of TTP to palm fruit and the explanation of Blockchain technology what it is used for in TTP. Lastly, we explore the similar applications in the market or even proposed by other researchers. The three applications that are compared are Google Lens, 9 Angle Crop and PaLife.

CHAPTER THREE

RESEARCH METHODOLOGY

In this chapter, the methodology used for the project is provided to help in creating a strategy on planning and developing the system. The software development methodology used for this project is the waterfall model which have 4 phases that describes the activities and outcomes done throughout the duration of the project development.

3.1 Waterfall Model

Waterfall model is a software development methodology that uses a linear and sequential approach that based on the software development lifecycle (Luthkevich, 2022). It is applied to projects that needs clear, consistent, and is unlikely to change during the course of development. This methodology originally consists of 6 phases but due to the time constraints of development for this project, it is decided to only apply 4 phases which are requirement gathering and analysis, system design, implementation, and testing. Figure 3.1 gives us a more detailed look at how the waterfall model structure look like.

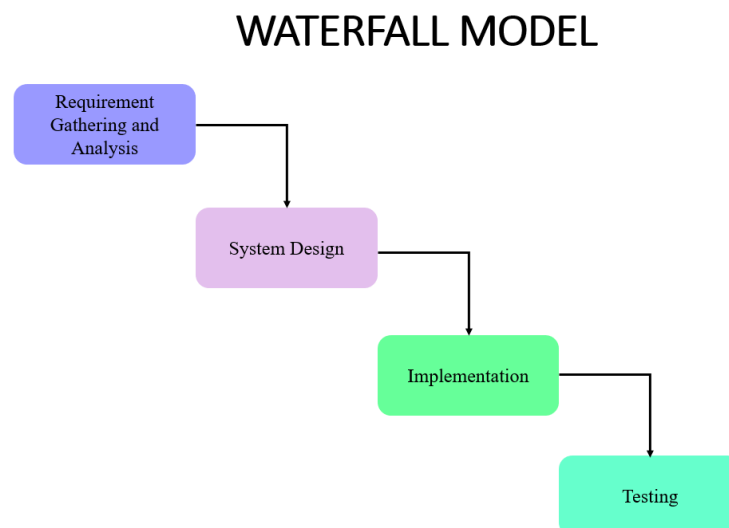


Figure 3.1 Waterfall Model Structure

(Adapted: tutorialspoint, 2022)

Phase	Activities	Outcome
Requirement Gathering and Analysis	• Discussing the system user identification • Discussing about user requirements	• System users • User requirements
System Design	Developing the design of: • System architecture • System flowchart • User case diagram • Use case description • User interface	• System architecture • System flowchart • User case diagram • Use case description • User interface
Implementation	• Discuss on hardware requirements used • Discuss on software requirements used • Developing fully function	• Knowing the hardware and software used to create the system • Develop the fully functional application
Testing	• Discuss the Test case	• Expected result

Table 3.1 Activities and Outcome of Waterfall Model

3.2 Requirement Gathering and Analysis

The requirement gathering and analysis is the first phase of the waterfall model. This phase is crucial in identifying and documenting the user requirements and expectations, which serve as the project's basis. In previous chapters, we have explored the identification of palm fruit ripeness which uses colour to determine whether the fruit is ripe or not. We also made comparison between different algorithm and similar applications where we pick the suitable ones to use for this project. With this we are able to answer the background of study, problem statement and objectives of the project.

3.2.1 System User Identification

Determining who the system users are is important because it helps us to underline who are the users of the system. This also helps in preparing what functions will be put into the system in order to fulfil the needs of the users.

1. Palm Fruit Farmers

- Responsible to harvest the palm fruits.
- Use their mobile phone to determine the ripeness of the palm fruit for the batches that arrive from the farm.
- Collect the palm fruits in batches and enter the information such as date of harvest, ripeness stage, where the batch comes from, and who harvested the palm fruits.
- Transfer the batches of palm fruit to the factory mill.

2. Factory Management Staff

Use the ripeness level from farmer and classify the weight and determine the quality of the palm fruit.

Send the palm fruit to the oil production section.

3.2.2 User Requirements

Determining the user requirements for the system is important because it reduce the possibility of misunderstandings or misinterpretations between the developers and users. This decreases the likelihood of producing a system that falls short of user expectations.

1. User Authentication

- The system will need a login page that allows the user to enter their credentials in order to access the functions of the system.

2. User Friendly Interface

- The system's interface should be simple and easy to use, allowing users to explore and interact with the functions with ease.

3. Generate Report

- The system should be able to generate reports of the batches of palm fruits during transportation to the mill and show the information which shows which batch is ripe and unripe.

4. Palm Fruit Identification

- The system should be able to have a function that allows the user to take a picture or image of a palm fruit to determine the ripeness stage.

5. Palm Fruit Quality Grading

- The system should be able to grade the quality for the oil production based on the weight and ripeness stage of the palm fruit.

6. Traceability

- The system should be able to trace the ripe batch of palm fruit that was sent to the mill. This will help in separating the good batches from the bad ones.

3.3 System Design

The system design is the second phase of the waterfall model. This phase will act as a blueprint where it will turn the requirements that is acquired from the previous phase into a thorough system design. This phase will show the system architecture, flowchart, use case diagram and interface design of the system.

3.3.1 System Architecture

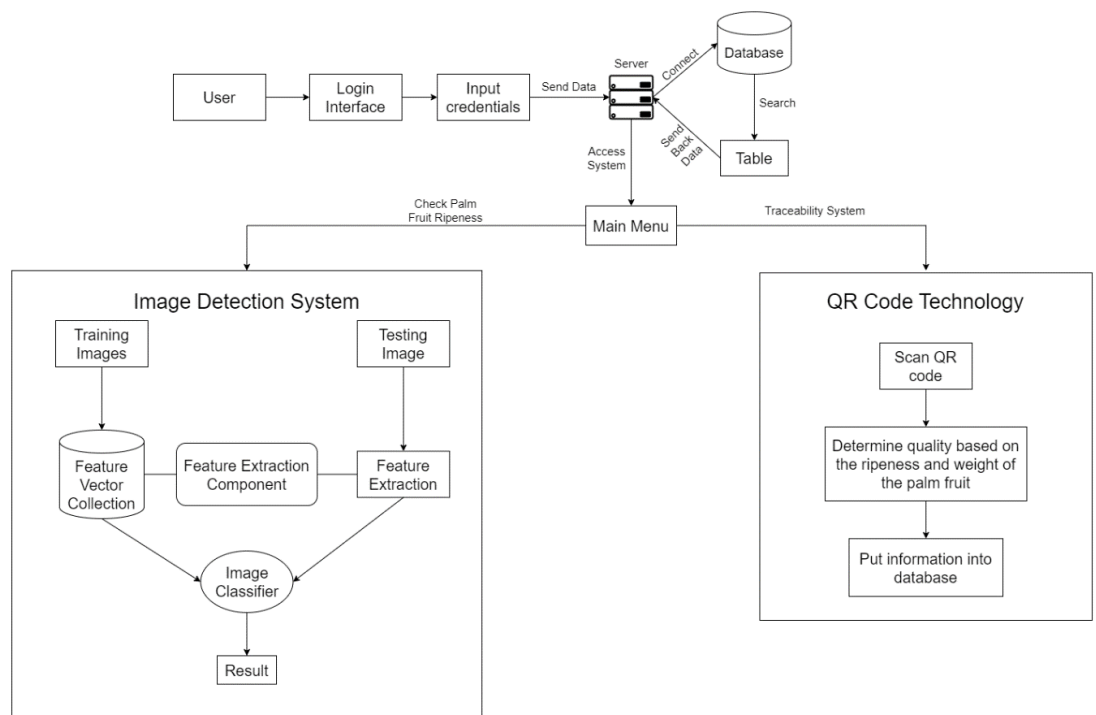


Figure 3.2 System Architecture of the project

(Adapted: Tran, 2016)

Based on Figure 3.2, the user will first enter the login interface where they will need to enter their credentials. After that, the data entered will be checked in the database if the data exist or not. When the data is found, the user will be able to access the homepage where they are able to check the ripeness of the palm fruit and use the traceability system.

In the image detection system architecture, the user will first submit an image into the system where it will be labelled as testing image. With the testing image, the system will use the training images to train the machine and compare with the testing

image and produce the results which shows if the palm fruit in the picture is ripe or not.

For the QR code technology system architecture, after scanning the QR code for the specific batch of palm fruits, the user will be able to determine the quality of the palm fruit produced based on the ripeness data that was taken from the farmer and the weight of the palm fruit.

3.3.2 System Flowchart

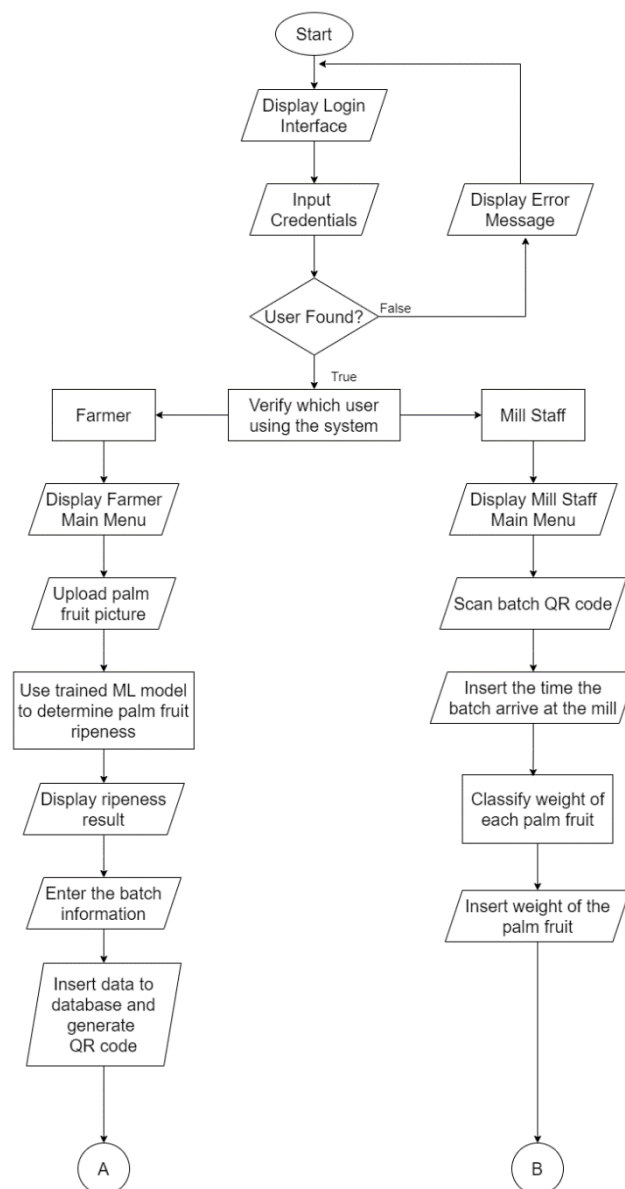


Figure 3.3 System Flowchart

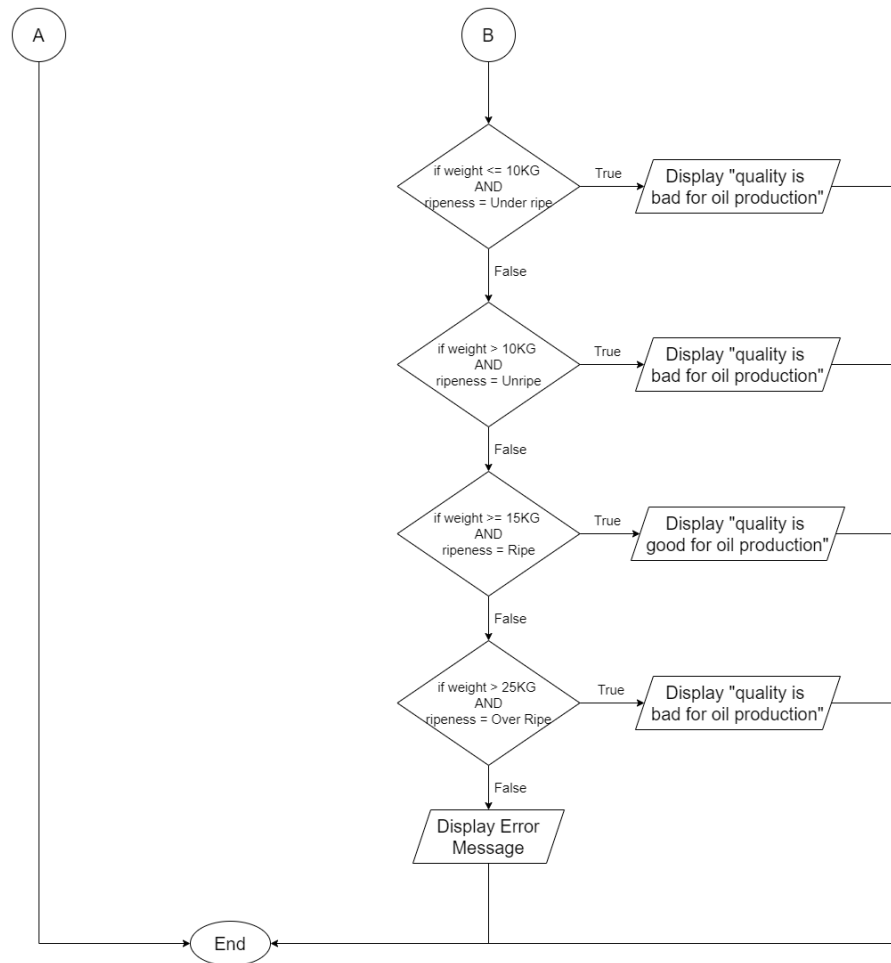


Figure 3.4 System Flowchart (Continued)

Based on Figure 3.3 and Figure 3.4, the system flowchart will start at the login page where the users will need to enter their credentials like username and password. If the credentials are true, the system will verify whether the user is a farmer or a staff.

If the user is verified as a farmer, they will be able to upload an image of a palm fruit and with that image the machine learning model will use the trained data to determine whether the palm fruit in the picture is ripe or not. After determining the stage of ripeness of the palm fruit, the farmer will then pick up the palm fruit and put it in batches. Here the farmer will enter important information like the ripeness of the palm fruit, the location of the farm, the date of when the palm fruit was harvest, and the farmer's information to know who harvested the palm fruit.

If the user is verified as a factory management staff, they will be able to see the information of the batches that arrived at the mill. They will do grading of the palm fruit depending on the weight and ripeness stated by the farmer in the system. Then the data

will be put inside the database.

3.3.3 Algorithm Design

This section shows the details of how the system will be used by the users in pseudocode form. It highlights the conditional statements that is used in system.

Algorithm 1: Login Process

```
STEP 1:  START
STEP 2:  Input Credentials
STEP 3:  IF user found
          THEN verify user
          IF user = "farmer"
              THEN Display Farmer Main Menu Interface
          ELSE IF user = "mill_staff"
              THEN Display Mill Staff Main Menu Interface
          ELSE
              Display error message
        ELSE
              Display error message and go back to Login Interface
```

Algorithm 2: Palm fruit ripeness detection

```
STEP 1:  Display Farmer Main Menu Interface
STEP 2:  Upload palm fruit image
STEP 3:  Load the trained ML model
STEP 4:  Preprocess the uploaded image
          Resize the image to the required input size of the model
STEP 5:  Put the pre-processed image into the ML model
STEP 6:  Get prediction from the ML model
          Extract the ripeness detection (under ripe, unripe, ripe or over ripe)
STEP 7:  Display the result
```

Algorithm 3: Determine the quality of the palm fruit batch

STEP 1: Display Mill Staff Main Menu Interface

STEP 2: Scan batch QR code

STEP 3: Insert the time the batch arrive at the mill

STEP 4: Insert the weight of each palm fruit

STEP 5: IF weight $\leq$ 10KG AND ripeness = Under ripe
 THEN Display “quality is bad for oil production”
ELSE IF weight $>$ 10KG AND ripeness = Unripe
 THEN Display "quality is bad for oil production"
ELSE IF weight $\geq$ 15KG AND ripeness = Ripe
 THEN Display "quality is good for oil production"
ELSE IF weight $\geq$ 25KG AND ripeness = Over Ripe
 THEN Display "quality is bad for oil production"
ELSE
 THEN Display error message

3.3.4 Entity Relationship Diagram (ERD)

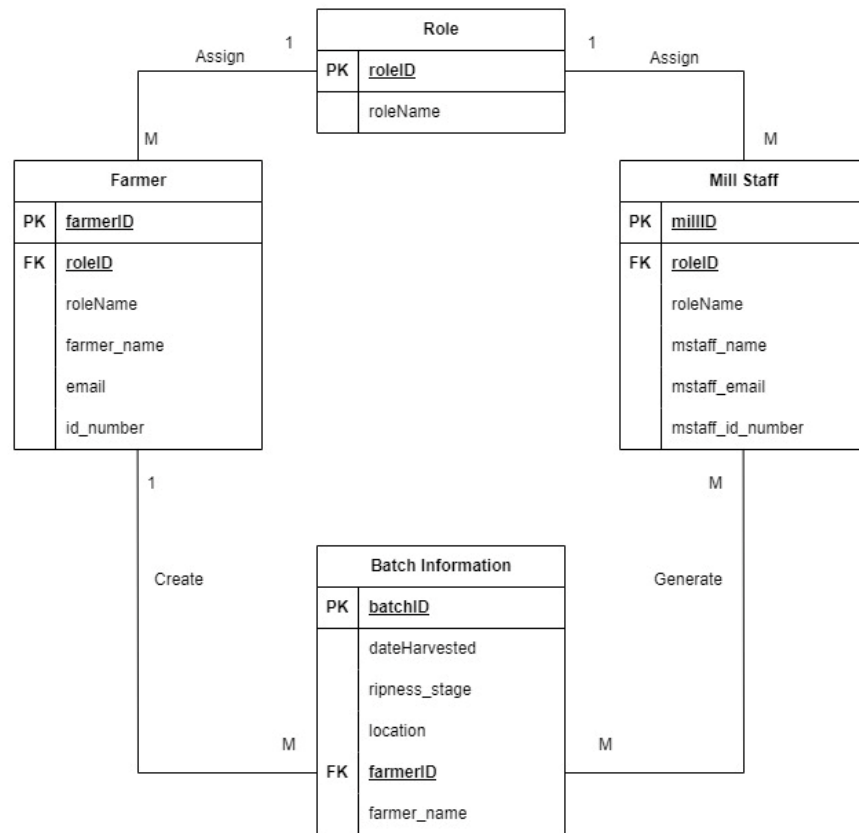


Figure 3.5 Entity Relationship Diagram

Based on Figure 3.5 the system ERD shows how the entities inside the database will connect with each other. The role table have an ID called roleID and an attribute called roleName. This table will contain the roles that the employees will be given based on their position. For example, the role name for farmers will be farmers while factory management staff will be named staff.

Next, we have the farmer table which contains an ID called farmerID and a foreign key that is linked to the role table. The attributes for farmers are their name, email, and id number. Aside from that, the staff table also contains an ID called millID and have a foreign key that is linked to the role table. It also has the attributes for staff which are name, email, and id number.

Lastly, the batch information table have an ID called batchID and a foreign key which is linked to the farmer table. This foreign key is needed in order to know which farmer has entered the information for the batch of palm fruit. For the farmer table, they

are able to insert data into the batch information table. For staff table, they will be able to see the batch information.

3.3.5 Use Case Diagram

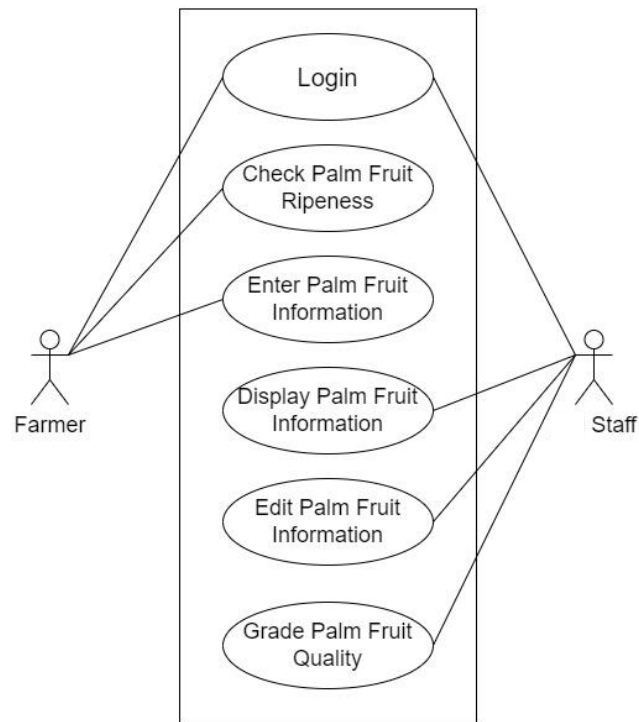


Figure 3.6 Use Case Diagram

1. Login
 - Provide secure login mechanism for farmer and staff.
 - Use credentials like username and password to login into the system.
2. Check Palm Fruit Ripeness
 - Farmers will upload a photo of a harvested palm fruit to determine its ripeness.
 - Show result of ripeness.
3. Enter Palm Fruit Information
 - Farmer will enter the harvested palm fruit's information like date harvested, location and ripeness stage into the database.

4. Display Palm Fruit Information

- Staff can display data of the inputted palm fruit information.

5. Edit Palm Fruit Information

- Staff will be able to edit the information of the harvested palm fruit.

6. Grade Palm Fruit Quality

- Staff will do the quality grading by weighting the palm fruit and use the ripeness stage taken by the farmer previously.

3.3.6 Use Case Description

Use Case Name	Login
Description	The farmer and staff will login to the system in order to use the system
Pre-Condition(s)	Farmer and staff already have an account
Post-Condition(s)	Depends on which user will be using the system, if the user is farmer the system will directed to the farmer main menu while staff will be directed to the staff main menu
Actor	Farmer, Staff
Flow of Events	
Primary Flow	• The user will enter their credentials such as username and password and press the login button. • The database will check and validate the user information. Once the credentials match, if the user is farmer the system will directed to the farmer main menu while staff will be directed to the staff main menu
Alternative / Exception Flow	If the user aborts the use case, they will return to the login page

Table 3.2 Login Use Case Description

Use Case Name	Check Palm Fruit Ripeness
Description	The farmer is able to identify the ripeness of the palm fruit by using their mobile phone.
Pre-Condition(s)	The farmer already has pictures of the palm fruit.
Post-Condition(s)	After uploading the picture into the system, it will use the trained machine learning model to compare and produce result.
Actor	Farmer
Flow of Events	
Primary Flow	• The farmer will upload pictures of palm fruit and press the identify button. • The machine learning model will determine the ripeness of the palm fruit. • Results of which stage is the palm fruit currently at will be shown.
Alternative / Exception Flow	Not applicable

Table 3.3 Check Palm Fruit Ripeness Use Case Description

Use Case Name	Enter Palm Fruit Information
Description	The farmer will be able to enter the information of the harvested batch of palm fruit
Pre-Condition(s)	The farmer will enter information of the palm fruit batches
Post-Condition(s)	Palm fruit information successfully inserted into the database
Actor	Farmer
Flow of Events	
Primary Flow	• The farmer will open the enter palm fruit information interface and enter some information about the harvested palm fruit. • The palm fruit information is recorded to the database.
Alternative / Exception Flow	Not applicable

Table 3.4 Enter Palm Fruit Information Use Case Description

Use Case Name	Display Palm Fruit Information
Description	The staff will be able to see the information of the palm fruit
Pre-Condition(s)	Staff is already logged in
Post-Condition(s)	Staff can see the palm fruit information
Actor	Staff
Flow of Events	
Primary Flow	• Staff will press the display palm fruit information. • The system will display all the information of the batches of palm fruit arrived.
Alternative / Exception Flow	Not applicable

Table 3.5 Display Palm Fruit Information Use Case Description

Use Case Name	Edit Palm Fruit Information
Description	• The staff is able to generate the full report of the palm fruit information
Pre-Condition(s)	Staff is already logged in
Post-Condition(s)	Staff generate the batch information that can be save into a file
Actor	Staff
Flow of Events	
Primary Flow	• The staff will press the generate palm fruit report. • The system will display the palm fruit report. • Show button to print or save into the device.
Alternative / Exception Flow	Not applicable

Table 3.6 Edit Palm Fruit Information Use Case Description

3.3.7 User Interface

User interface is important for the system design because it can help to get feedback from users and end users can be obtained early in the development phase by presenting the UI design. This makes sure that before making big developments, their preferences and expectations are considered. Aside from that, potential design flaws, consistency problems, or usability issues can be found early in the process by displaying the UI design. By resolving these problems before to the project's actual implementation, this lessens the possibility of expensive adjustments being made later on.

1. Login Interface

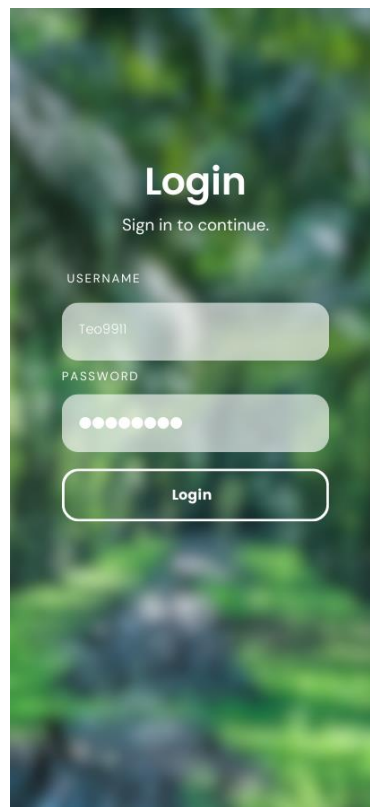


Figure 3.7 Login Interface

Figure 3.7 shows the interface for the login page. Users of the system will need to use their username and password to enter the mobile app.

2. Farmer Main Menu Interface



Figure 3.8 Farmer Main Menu Interface

After login, if the system detects that the user logging in is a farmer, they will be directed to the farmer main menu. Here the farmer can press the “Check Ripeness” to check the ripeness of the harvested palm fruit. Aside from that, they also need to insert the palm fruit harvested data into the system.

3. Ripeness Detection Interface

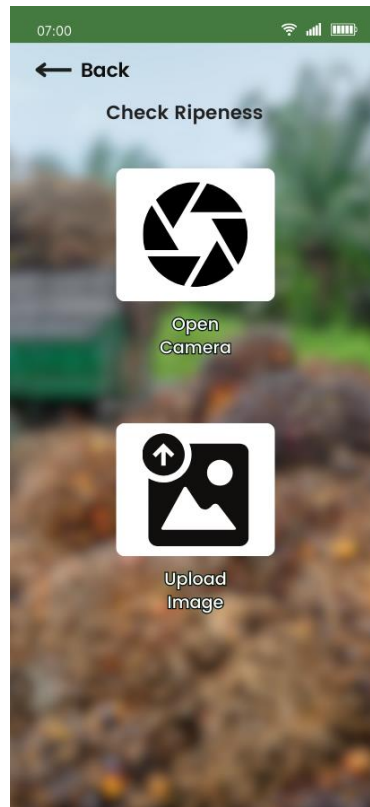
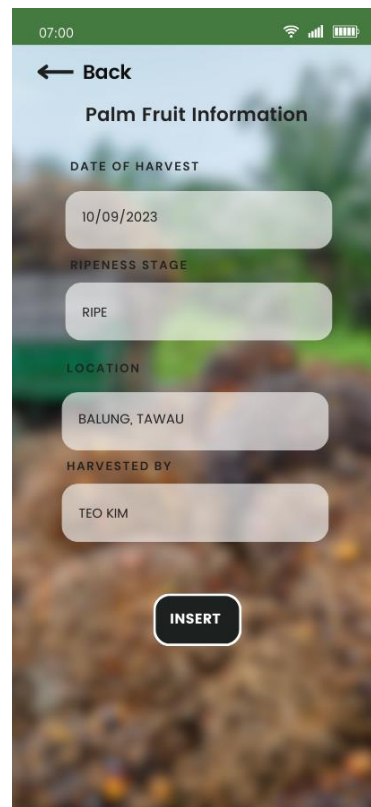


Figure 3.9 Ripeness Detection Interface

From the farmer main menu, when clicking the Check Ripeness button, the farmer will be directed to the Ripeness Detection Interface where the farmer will have two buttons which are “Open Camera” which opens the camera to allow them to take picture of the palm fruit or “Upload Image” which opens the gallery to allow them to upload a picture of palm fruit.

4. Enter Palm Fruit Information Interface



The screenshot shows a mobile application interface for entering palm fruit information. At the top, there is a green status bar with the time 07:00, signal strength, and battery level. Below this is a white header bar with a back arrow and the text "Back". The main title "Palm Fruit Information" is centered below the header. The form consists of five input fields, each with a label above it: "DATE OF HARVEST" with the value "10/09/2023", "RIPENESS STAGE" with the value "RIPE", "LOCATION" with the value "BALUNG, TAWAU", and "HARVESTED BY" with the value "TEO KIM". Each input field is a white rounded rectangle with a light gray border. At the bottom of the form is a black rounded rectangle with the text "INSERT" in white. The background of the app is a blurred image of palm fruit clusters.

Figure 3.10 Enter Palm Fruit Information Interface

From the farmer main menu, when clicking the Enter Palm Fruit Information button, the farmer will be directed to the Palm Fruit Information Interface where the farmer in charge of harvesting the palm fruits will insert data of the palm fruit harvested.

5. Staff Main Menu Interface

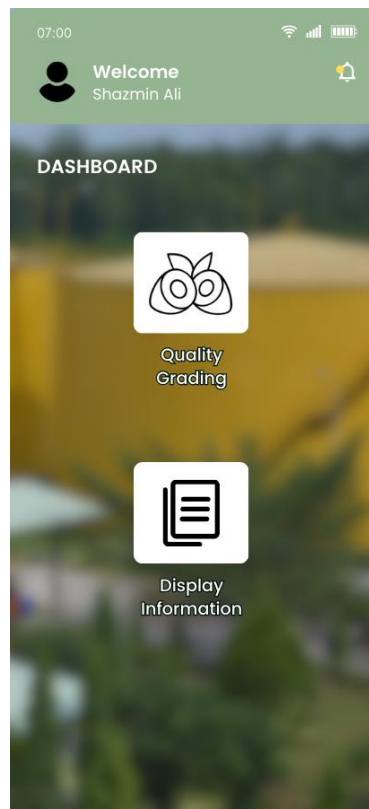
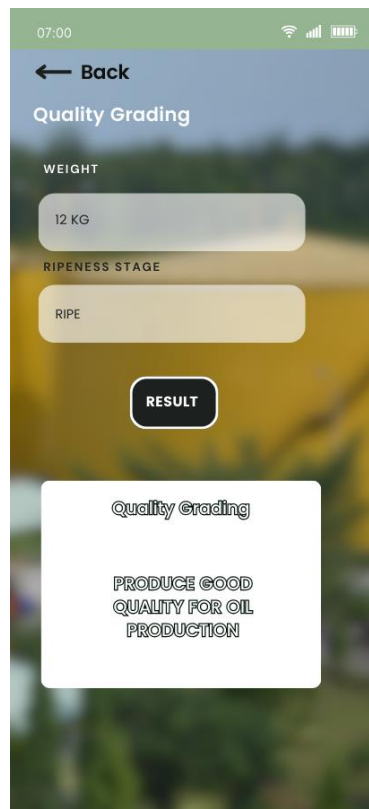


Figure 3.11 Staff Main Menu Interface

After login, if the system detects that the user logging in is a staff, they will be directed to the Staff Main Menu. Here the staff can press the “Quality Grading” to grade the palm fruit arrive to the mill. Aside from that, they also can display the information of the graded palm fruit.

6. Quality Grading Interface



The screenshot shows a mobile application interface for quality grading. At the top, there is a status bar with the time 07:00 and signal indicators. Below this is a navigation bar with a back arrow and the text "Back". The main title "Quality Grading" is displayed. Under the title, there are two input fields: "WEIGHT" with the value "12 KG" and "RIPENESS STAGE" with the value "RIPE". Below these fields is a button labeled "RESULT". At the bottom, there is a white box containing the text "Quality Grading" and "PRODUCE GOOD QUALITY FOR OIL PRODUCTION".

Figure 3.12 Quality Grading Interface

From the Staff Main Menu, when pressing the Quality Grading button, the staff will be directed to the Quality Grading Interface where the staff will insert the weight and ripeness stage of the palm fruit. After pressing the Result button, a text will appear that tells whether the palm fruit is good or bad for oil production.

7. Palm Fruit Information Interface

HARVEST DATE	RIPENESS STAGE	LOCATION	HARVESTED BY

GENERATE REPORT

Figure 3.13 Palm Fruit Information Interface

From the Staff Main Menu, when pressing the Palm Fruit Information button, the staff will be directed to the Palm Fruit Information Interface where the staff will be able to see the information of the graded palm fruit.

3.4 System Implementation

The implementation phase is the third phase of the waterfall model. This phase will show the hardware and software requirements in order to develop and run the project. It follows the design phase in order to show how will the system be developed. It also explains the ways in which the system will be implemented.

3.4.1 Hardware Requirements

Hardware requirements refers to the specifications of device components that is needed to run the system. It is important to identify the requirements for the hardware in order for the system to run efficiently and effectively. Here are some suggested requirements that is needed to create a mobile application.

Hardware	Requirements
Memory	At least 8GB RAM
Disk Space	At least 8GB of storage
Graphics Processing Unit (GPU)	NVIDIA GeForce MX330
Operating System	Windows 10 64-bit

Table 3.7 Hardware Requirements

To develop the mobile application, having sufficient memory is important to run the software that will be used. That is why having the memory to have at least 8GB of RAM helps to run the software like Android Studio smoothly. Aside from that, having at least 8GB of storage will makes sure to have enough storage to save project files, install and maintain the program, and provide the space needed for system updates and temporary files.

The GPU will help to improve the performance of tasks that involves with graphics. This can be especially helpful when using applications with a lot of visuals or when working with intricate user interface elements. Lastly, to create the mobile application, windows 10 will be used as the Operating System.

3.4.2 Software Requirements

Software requirements refers to the specifications that defines the functions, features, and characteristics of what the system will do. It will serve as a starting point for the system's design, development, and testing. Here are some of the software and programming languages that will be used for the creation of the mobile application.

Software

Software	Function
Android Studio	An Integrated Development Environment (IDE) software to create android applications
Tensorflow and Keras	Acts as a software library to develop machine learning systems
Visual Studio Code	A code editor software to write machine learning system
Canva	An online graphic design tool to create the design of the mobile application

Table 3.8 Software Requirements

Programming Language

Programming Language	Function
Java	Used in android studio to write the functionality of the android app
Python	Used in Tensorflow and Keras for machine learning

Table 3.9 Programming Language Used

The software that will be used for development of the mobile application is Android Studio. It is an IDE software that is commonly used for android studio. Java will also be used in Android Studio to write the functions that will interact between the application.

Next, Tensorflow and Keras will be used for the development of the image detection of the palm fruit. TensorFlow is an open-source machine learning framework developed by Google. Keras, on the other hand, is a high-level neural networks API that

runs on top of TensorFlow. Together, they provide a powerful toolkit for building and training machine learning models. To run these, Visual Studio Code will be used. It is a code editor software that is used to write many programs.

3.5 Testing

3.5.1 Test Case

Interface 1: Login (Farmer and Staff)

Test ID	Test Description	Preconditions	Actions	Expected Results
TEST1	Login function with valid credentials for farmers	Farmer already have an account	Enter the valid username and password. Press the login button	The system will connect to the database and validates the credentials. The farmer is logged into the system.
TEST2	Login function with invalid credentials for farmers	Farmer and staff already have an account	Enter the invalid username and password. Press the login button	System will display an error message. Farmer is not logged into the system.
TEST3	Login function with valid credentials for staff	Staff already have an account	Enter the valid username and password. Press the login button	The system will connect to the database and validates the credentials. The staff is

				logged into the system.
TEST4	Login function with invalid credentials for staff	Staff already have an account	Enter the invalid username and password. Press the login button	System will display an error message. Staff is not logged into the system.

Table 3.10 Login Test Case

Interface 2: Palm Fruit Ripeness Detection

Test ID	Test Description	Preconditions	Actions	Expected Results
TEST5	The farmer will check the ripeness of the palm fruit	The farmer already has pictures of the palm fruit	Farmer will upload into the system. The Machine Learning Model will use the testing image and compare with the trained model to determine the ripeness.	Display the stage of ripeness of palm fruit.

Table 3.11 Palm Fruit Ripeness Detection Test Case

Interface 3: Enter Palm Fruit Information

Test ID	Test Description	Preconditions	Actions	Expected Results
TEST6	The farmer will enter the information of the palm fruit batch harvested	The farmer is already logged into the system.	Farmer will put in the information of the harvested batch of palm fruit.	Data is successfully entered into the database.

Table 3.12 Enter Palm Fruit Information Test Case

Interface 4: Display Palm Fruit Information and Generate Report

Test ID	Test Description	Preconditions	Actions	Expected Results
TEST7	The information of the harvested palm fruit will be shown for staff	The staff is already logged into the system.	Staff will press the display palm fruit information	Show the information of the harvested palm fruit.
TEST8	Staff is able to generate the report of the palm fruit information	The staff is already logged into the system.	The staff will press the generate palm fruit information report	The system will display the palm fruit information

Table 3.13 Display Palm Fruit Information and Generate Report Test Case

Interface 5: Grade Palm Fruit Quality

Test ID	Test Description	Preconditions	Actions	Expected Results
TEST9	Quality grade is not good for oil production	The staff is already logged into the system.	The staff will insert the weight of the palm fruit into the system	Show result of palm fruit grade as not good for oil production
TEST10	Quality grade is good for oil production	The staff is already logged into the system.	The staff will insert the weight of the palm fruit into the system	Show result of palm fruit grade as good for oil production

Table 3.14 Grade Palm Fruit Quality Test Case

3.6 Gantt Chart

A Gantt chart is a visual representation of a project schedule that shows the start and finish dates of various elements of the project. It is used to show the timeline of the project's activities taken place.

Semester 5

Week															
Activities	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Contact Potential Supervisors															
F1 (Mutual Acceptance Form)															
Chapter 1 Presentation															
F2 (Project Motivation Evaluation Form)+ Presentation slides															
Chapter 1															
Chapter 2 + F3 (Literature Review Evaluation Form)															
Chapter 3 + F4 (Methodology Evaluation Form)															
Submission of Full Report															
Final Proposal Presentation															

Figure 3.14 Semester 5 Gantt Chart

Semester 6

Week														
Activities	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Submission of Chapter 1 (Refined)														
Submission of Chapter 2 (Refined)														
Submission of Chapter 3 (Refined)														
Submission of Business Model Canvas														
Submission of Chapter 4														
Progress Presentation														
Submission of Chapter 5														
Submission of First Draft														
Submission of Full Report														
Presentation of Final Year Project														
Plagiarism Checking														
Development of Mobile Application														

Figure 3.15 Semester 6 Gantt Chart

3.7 Summary

This chapter covers the software development methodology used which is the Waterfall Model for its linear and sequential approach. In the waterfall model there are a total of 4 phases used for this project. The first phase is the requirement gathering and analysis phases which discusses on the system user identification and user requirements. The second phase is the system design where it explores the system architecture, system flowchart, entity relationship diagram (ERD), use case diagram, use case description and user interface. The third phase is the system implementation where it discusses the hardware requirements and software requirements. The last phase is the testing phase where it discusses the test case for the project.

CHAPTER FOUR

RESULTS AND DISCUSSION

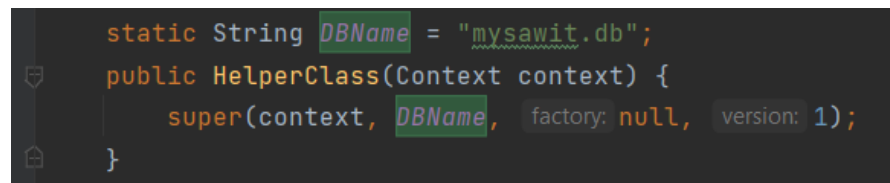
This chapter covers the testing and results of the developed mobile application where the project implementation and the output of the application is shown. Next, the results of the will be discussed in this chapter to evaluate the expected outcome of the project and the actual results of the completed project.

4.1 SQLite Database

This mobile application uses SQLite Database to store the farmer and staff's login information as well as the information of the palm fruit batched that has been harvested.

4.1.1 Database and Table Creation

The database is declared in a class file called HelperClass, where all of the functions such as inserting, updating, and deleting data is stored. It serves as the class that will connect the mobile application with the database.



```
static String DBName = "mysawit.db";
public HelperClass(Context context) {
    super(context, DBName, factory: null, version: 1);
}
```

Figure 4.1 Coding Segment to Create Database in SQLite

Figure 4.1 shows the creation of the HelperClass where it takes a context object as the parameter where it provides access to the resources and information in the Android Studio. The HelperClass calls out the super constructor and declare the creation of the database by using the DBName variable which is a string that contains the name of the database for the mobile application.

```

String createTableUser = "CREATE TABLE 'users' ('userid' INTEGER NOT NULL, 'fullname' TEXT NOT NULL, 'username' TEXT NOT NULL, " +
    "'email' TEXT NOT NULL, 'phoneNo' TEXT NOT NULL, 'password' TEXT NOT NULL, 'accessid' INTEGER NOT NULL, " +
    "'PRIMARY KEY('userid' AUTOINCREMENT), FOREIGN KEY('accessid') REFERENCES 'userAccess'('accessid') ON DELETE SET NULL ON UPDATE CASCADE)";

String createUserAccess = "CREATE TABLE 'userAccess' ('accessid' INTEGER NOT NULL, 'usertype' TEXT NOT NULL, PRIMARY KEY('accessid'))";

String createBatchInfo = "CREATE TABLE 'batchInfo' ('batchid' INTEGER NOT NULL, 'ripenessStage' TEXT, 'dateHarvest' TEXT NOT NULL, " +
    "'location' TEXT NOT NULL, 'qr' BLOB, PRIMARY KEY('batchid' AUTOINCREMENT))";

String createQualityProduction = "CREATE TABLE 'qualityProduction' ('qualityid' INTEGER NOT NULL, 'weight' INTEGER NOT NULL, " +
    "'batchid' INTEGER, 'result' TEXT, PRIMARY KEY('qualityid' AUTOINCREMENT), FOREIGN KEY('batchid') REFERENCES 'batchInfo')";

@Override
public void onCreate(SQLiteDatabase sqLiteDatabase) {
    sqLiteDatabase.execSQL(createTableUser);
    sqLiteDatabase.execSQL(createUserAccess);
    sqLiteDatabase.execSQL(createBatchInfo);
    sqLiteDatabase.execSQL(createQualityProduction);
}

@Override
public void onUpgrade(SQLiteDatabase sqLiteDatabase, int i, int ii) {
    sqLiteDatabase.execSQL("DROP TABLE IF EXISTS users");
    sqLiteDatabase.execSQL("DROP TABLE IF EXISTS userAccess");
    sqLiteDatabase.execSQL("DROP TABLE IF EXISTS batchInfo");
    sqLiteDatabase.execSQL("DROP TABLE IF EXISTS qualityProduction");
    onCreate(sqLiteDatabase);
}

```

Figure 4.2 Coding Segment to Create Tables in SQLite

Figure 4.2 shows the creation of all the necessary tables that will be used in the mobile application. It firstly declares the table name in a string for each table and assigning it with the SQLite coding to create the tables. The tables that are created are users table, user access table, batch information table and quality production table. After declaring the tables, it will be put inside an onCreate method to create the table. The sqLiteDatabase parameter allows the method to execute SQL commands on the database while the sqLiteDatabase.execSQL() method is used to insert the table to the database. After inserting the table, the application will check whether the table has already been created. If the table has already been created, the database will drop the table that was created in the previous method.

4.1.2 Data Insertion and Retrieval

```

inputDataBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        String ripenessStage = result.getText().toString();
        String dateHarvest = dateBtn.getText().toString();
        String location = locationSpinner.getSelectedItem().toString();

        if(result.length() > 1 && dateBtn.length() > 1 && !location.equals("Please select a location")){
            ContentValues cv = new ContentValues();
            cv.put("ripenessStage", ripenessStage);
            cv.put("dateHarvest", dateHarvest);
            cv.put("location", location);

            dbHelper.insertBatchInfo(cv);

            String recentbatchid = recentid();

            generateAndSaveQRCode(Integer.parseInt(recentbatchid));

            Toast.makeText(context, addBatchInfo.this, "Data Inserted Successfully!", Toast.LENGTH_SHORT).show();

            Intent intent = new Intent(context, addBatchInfo.this, ripeness_detection.class);
            startActivity(intent);
        }
        else{
            Toast.makeText(context, addBatchInfo.this, "Please fill in the informations!", Toast.LENGTH_SHORT).show();
        }
    }
}

```

Figure 4.3 Coding Segment to Insert the Palm Fruit Batch Information

```

public void insertBatchInfo(ContentValues contentValues){
    getWritableDatabase().insert( table: "batchInfo", nullColumnHack: "", contentValues);
}

```

Figure 4.4 Coding Segment to Insert Palm Fruit Batch Information in HelperClass

In Figure 4.3, it demonstrates inserting the data to the database by using the ContentValues object. It firstly validates to see whether the text fields are empty. If it is not empty, the ContentValues object is used to keep the data entered by the user. Then the data is inserted into the database using the insertBatchInfo() method found in the HelperClass class as shown in Figure 4.4.

4.1.3 Generating QR Code

```

private void generateAndSaveQRCode(int text) {
    MultiFormatWriter writer = new MultiFormatWriter();
    try {
        BitMatrix matrix = writer.encode(String.valueOf(text), BarcodeFormat.QR_CODE, width: 400, height: 400);
        BarcodeEncoder encoder = new BarcodeEncoder();
        Bitmap bitmap = encoder.createBitmap(matrix);

        // Convert Bitmap to byte array
        ByteArrayOutputStream stream = new ByteArrayOutputStream();
        bitmap.compress(Bitmap.CompressFormat.PNG, quality: 100, stream);
        byte[] byteArray = stream.toByteArray();

        // Save byte array to SQLite database
        dbHelper.updateBatchInfoQRCode(String.valueOf(text), byteArray);
    } catch (WriterException e) {
        e.printStackTrace();
    }
}

```

Figure 4.5 Coding Segment to Generate QR Code

In Figure 4.5, it shows the coding segment to generate a QR code after the batch information was created. By using the ZXing library, a QR code image is generated where the integers are converted to string as its content. After the QR code image is generated with the size of 400 pixel for height and width in Bitmap, it will then convert to a byte array by compressing it to a PNG format for easier access. Then, the QR code will be updated to the palm Fruit batch information table with the recent data inserted to the table.

4.2 Ripeness Detection Implementation

```
public void classifyImage(Bitmap image){
    try {
        PalmFruitRipenessDetectionModelVersion2 model = PalmFruitRipenessDetectionModelVersion2.newInstance(getApplicationContext());

        // Creates inputs for reference.
        TensorBuffer inputFeature0 = TensorBuffer.createFixedSize(new int[]{1, 256, 256, 3}, DataType.FLOAT32);
        ByteBuffer byteBuffer = ByteBuffer.allocateDirect(4 * imageSize * imageSize * 3);
        byteBuffer.order(ByteOrder.nativeOrder());

        int[] intValues = new int[imageSize * imageSize];
        image.getPixels(intValues, 0, image.getWidth(), 0, 0, image.getWidth(), image.getHeight());
        int pixel = 0;
        for(int i=0; i < imageSize; i++){
            for(int j = 0; j < imageSize; j++){
                int val = intValues[pixel++]; //RGB
                byteBuffer.putFloat(((val >> 16) & 0xFF) * (1.f / 255));
                byteBuffer.putFloat(((val >> 8) & 0xFF) * (1.f / 255));
                byteBuffer.putFloat((val >> 0) & 0xFF) * (1.f / 255);
            }
        }

        inputFeature0.loadBuffer(byteBuffer);

        // Runs model inference and gets result.
        PalmFruitRipenessDetectionModelVersion2.Outputs outputs = model.process(inputFeature0);
        TensorBuffer outputFeature0 = outputs.getOutputFeature0AsTensorBuffer();

        float[] confidences = outputFeature0.getFloatArray();
        //Find index with biggest confidence
        int maxPos = 0;
        float maxConfidence = 0;
        for(int i=0; i<confidences.length; i++){
            if(confidences[i] > maxConfidence){
                maxConfidence = confidences[i];
                maxPos = i;
            }
        }

        String[] classes = {"Overripe", "Ripe", "Under ripe", "unripe"};
        result.setText(classes[maxPos]);

        String s = "";
        for(int i = 0; i < classes.length; i++){
            s += String.format("%s: %.1f%%\n", classes[i], confidences[i] * 100);
        }
        confidence.setText(s);

        // Releases model resources if no longer used.
        model.close();
    } catch (IOException e) {
        // TODO Handle the exception
    }
}
```

Figure 4.6 Coding Segment for Ripeness Detection Implementation

In Figure 4.6, it shows the implementation for the palm fruit ripeness detection to the mobile application. The model is firstly loaded into the application by initializing the instance of `PalmFruitRipenessDetectionModelVersion2`. Then, prepare the input data by creating a `TensorBuffer` and `ByteBuffer` to process the machine learning capabilities of the model that uses `Tensorflow`. After that, the image that will be used to be identified needs to be processed by setting the properties of the image such as the weight and height. Next, the model will be run, and the process of the model's output is done which will retrieve the confidence level of the model. Finally, the result of the detection is done by detecting one out of the four categories of the ripeness stages which are over ripe, ripe, under ripe or unripe.

4.3 Application Interface

In this section, the interface of the mobile application is shown as well as the features that is implemented to the project.

4.3.1 Log in Interface

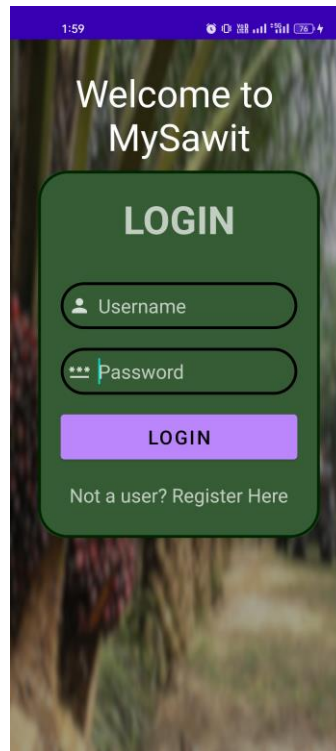


Figure 4.7 Login Interface

Figure 4.7 shows the login interface where both farmer and staff will be able to log into the system by inserting their username and password. If the credentials are correct, both farmer and staff will be able to enter the system based on their user access level. However, if the credentials are wrong an error message will appear to inform that the username or password is incorrect.

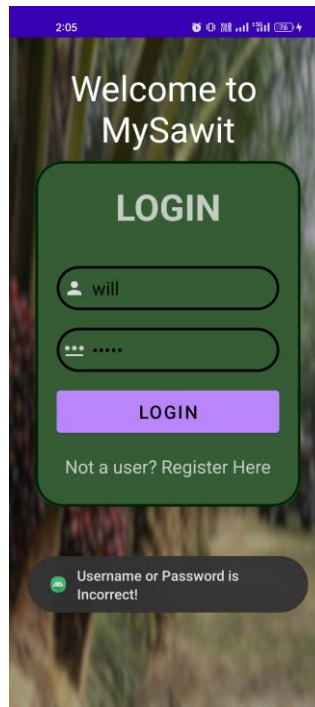


Figure 4.8 Error Message of Incorrect Credentials

4.3.2 Register Interface

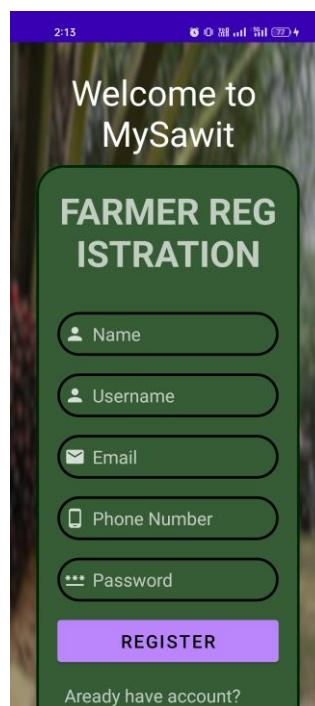


Figure 4.9 Farmer Registration

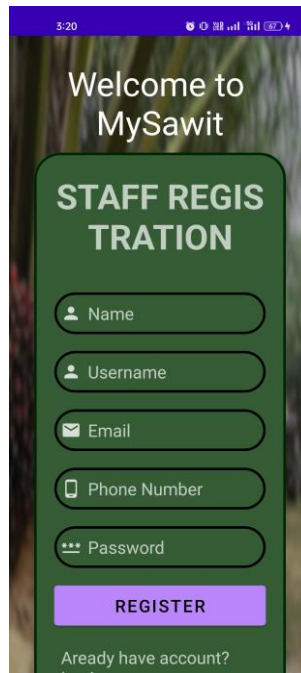


Figure 4.10 Staff Registration

Figure 4.9 and Figure 4.10 shows the registration interface for both users. They will be required to enter their name, username, email address, phone number and password. After filling in the information they will click the register button and the data will be inserted into the database and they will be redirected to the login interface again.

4.3.3 Check Ripeness



Figure 4.11 Check Ripeness Interface

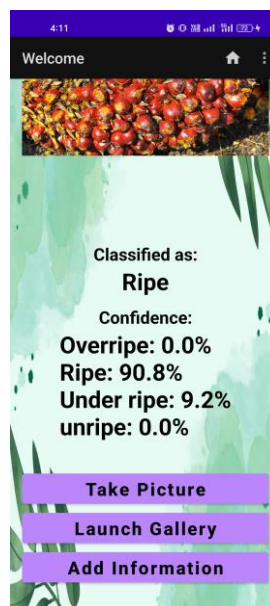
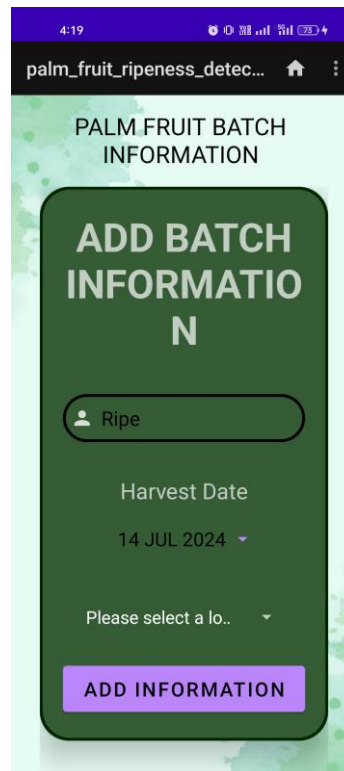


Figure 4.12 Ripeness Successfully Detected

In Figure 4.11 shows the interface to check the ripeness of palm fruit batches. It consists of the take picture button where the farmer will use their mobile phone's camera to capture the images of palm fruit. The farmers are also able to use their gallery to upload

pictures of palm fruit to determine the ripeness for the palm fruit. After the result comes out, the ripeness and confidence level of the detection is display. When clicking the add information button, farmers will be redirected to the add batch interface where the ripeness stage will be displayed.

4.3.4 Add Palm Fruit Batch Information



The screenshot shows a mobile application interface for adding batch information. At the top, there is a status bar with the time 4:19 and various icons. Below the status bar is a dark header with the text 'palm_fruit_ripeness_detec...' and a home icon. The main content area has a light green background with a darker green rounded rectangle in the center. Inside this rectangle, the text 'ADD BATCH INFORMATION' is displayed in large, bold, white letters. Below this text, there are three input fields: a dropdown menu with 'Ripe' selected, a text field for 'Harvest Date' with '14 JUL 2024' and a dropdown arrow, and another dropdown menu with the text 'Please select a lo..'. At the bottom of the rounded rectangle is a purple button with the text 'ADD INFORMATION' in white.

Figure 4.13 Add Batch Information Interface

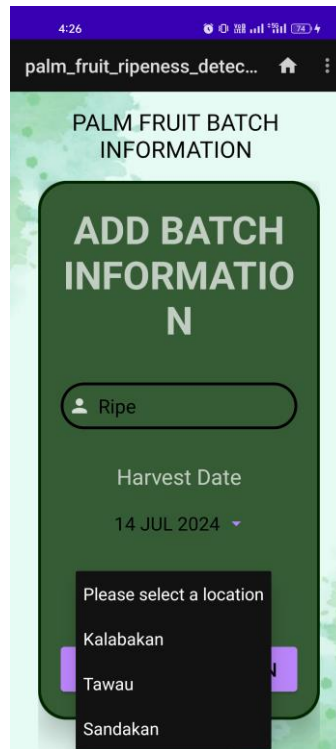


Figure 4.14 Dropdown Menu with Location

In Figure 4.13, the interface to add the batch information to the SQLite database is shown. The ripeness stage was taken from the ripeness detection interface and farmers cannot edit the result. Next, the harvest date is used to know when the palm fruit was harvested, and a dropdown menu is provided to determine where the palm fruit batch came from as seen In Figure 4.14.

4.3.5 Palm Fruit Batch Information

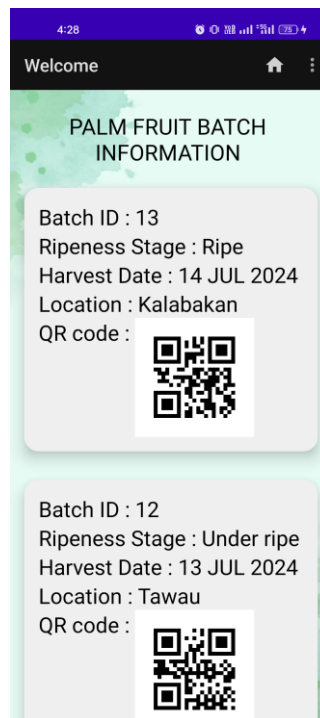


Figure 4.15 Palm Fruit Batch Information Interface

In Figure 4.15, all of the data is shown. The data displayed consist of batched, ripeness stage, harvest date and location. The QR code image generated is also added to the SQLite Database.

4.3.6 Quality Grading Interface

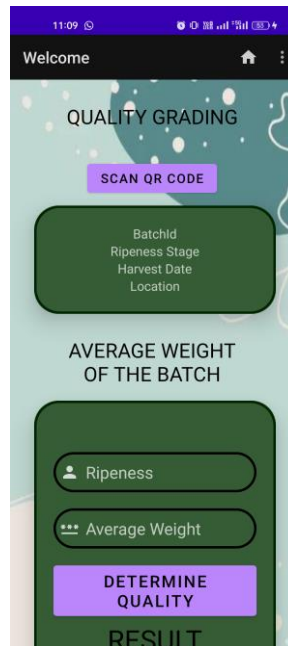


Figure 4.16 Quality Grading Interface

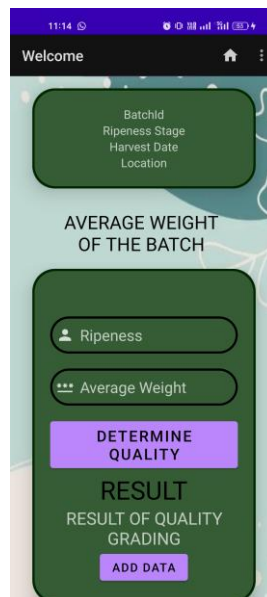


Figure 4.17 Determine the Quality of Palm Fruit

Figure 4.16 shows the quality grading interface that is only accessible to the staff working at the palm Fruit mill. The staff will firstly scan the QR code of the palm fruit batch that arrive to the mill to show the information of each batch. After that, pressing the determine quality button will display the result below of either the batch is suitable for oil production or have other uses such as become animal feed or plant fertilizer.

Then the staff will press the add data button to put the result inside the database.

4.3.7 Grading Information Interface

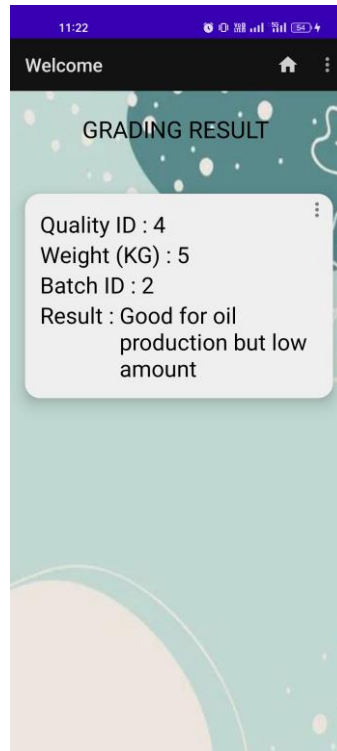


Figure 4.18 Grading Information Interface

Figure 4.18 shows the grading information that was inserted into the database. It displays the quality id, weight of the batch, the id of the batch scanned and the result of grading.

4.4 Functionality and Results

To test the functionality and see the results of the application, a test case is conducted to document the expected results and actual results as well as the status of the testing. The test description, preconditions and user actions are also included in the table below for accurate testing.

4.4.1 User Login Testing

Test ID	Test Description	Preconditions	Actions	Expected Results	Actual Result	Status
TEST1	Login function with valid credentials for farmers	Farmer already have an account	Enter the valid username and password. Press the login button	The system will connect to the database and validates the credentials. The farmer is logged into the system.	Farmer can log into the system successfully	Successful
TEST2	Login function with invalid credentials for farmers	Farmer and staff already have an account	Enter the invalid username and password. Press the login button	System will display an error message. Farmer is not logged into the	System display error message and farmer is redirected back to login page	Successful

				system.		
TEST3	Login function with valid credentials for staff	Staff already have an account	Enter the valid username and password. Press the login button	The system will connect to the database and validates the credentials. The staff is logged into the system.	Staff can log into the system successfully	Successful
TEST4	Login function with invalid credentials for staff	staff already have an account	Enter the invalid username and password. Press the login button	System will display an error message. staff is not logged into the system.	System display error message and staff is redirected back to login page	Successful

Table 4.1 User Login Functionality

Table 4.1 shows the testing of both farmer and staff logging into the system. Based on the results, both farmer and staff are able to log into the system when entering the right credentials and will show an error message when entering the wrong credentials.

4.4.2 Palm Fruit Ripeness Detection Testing

Test ID	Test Description	Pre conditions	Actions	Expected Results	Actual Results	Status
TEST5	The farmer will check the ripeness of the palm fruit	The farmer already has pictures of the palm fruit	Farmer will upload into the system. The Machine Learning Model will use the testing image and compare with the trained model to determine the ripeness.	Display the stage of ripeness of palm fruit.	The ripeness is correctly predicted and shown	Successful

Table 4.2 Palm Fruit Ripeness Detection Functionality

Table 4.2 shows the testing and functionality of the machine learning aspect of this application which is detecting the ripeness of the palm fruit. Based on the results the application is able to identify at which stage the ripeness of the palm fruit is.

4.4.3 Insert Batch Information Testing

Test ID	Test Description	Pre conditions	Actions	Expected Results	Actual Results	Status
TEST6	The farmer will enter the information of the palm fruit batch harvested	The farmer is already logged into the system.	Farmer will put in the information of the harvested batch of palm fruit.	Data is successfully entered into the database.	The data is successfully inserted to the database	Successful

Table 4.3 Insert Palm Fruit Information Functionality

Table 4.3 shows the functionality and testing of inserting information of the palm fruit batch to the database. Based on the results the data was successfully inserted to the database.

4.4.4 Displaying Palm Fruit Batch Information Testing

Test ID	Test Description	Preconditions	Actions	Expected Results	Actual Results	Status
TEST7	The information of the harvested palm fruit will be shown for staff	The staff is already logged into the system.	Staff will press the display palm fruit information	Show the information of the harvested palm fruit.	Harvested palm fruit is displayed	Successful
TEST8	Staff is able to generate the report of	The staff is already logged into	The staff will press the generate	The system will display the palm	Palm fruit information displayed	Successful

	the palm fruit information	the system.	palm fruit information report	fruit information		
--	----------------------------------	-------------	-------------------------------------	----------------------	--	--

Table 4.4 Display Palm Fruit Information

Table 4.4 show the functionality of displaying the Palm Fruit Batch Information to the staff. This allows them to see the records recorded by the farmers of the harvested batches.

4.4.5 Quality Grading of the Palm Fruit Batch Testing

Test ID	Test Description	Pre conditions	Actions	Expected Results	Actual Results	Status
TEST 9	Quality grade is not good for oil production	The staff is already logged into the system.	The staff will insert the weight of the palm fruit into the system	Show result of palm fruit grade as not good for oil production	Results show graded palm fruit not good for oil production	Successful

TEST 10	Quality grade is good for oil production	The staff is already logged into the system.	The staff will insert the weight of the palm fruit into the system	Show result of palm fruit grade as good for oil production	Results show graded palm fruit good for oil production	Successful
------------	---	---	--	--	--	------------

Table 4.5 Determine Quality of Palm Fruit

Table 4.5 shows the testing and functionality of determining quality of the oil produced by the batches.

4.5 Discussions

During the development of the application there were a number of difficulties faced. One of it was developing the application using Android Studio due to its high use of resources for the software to run. With a device that fits only the minimal requirements for it to run is difficult due to the device frequently being slow when running the android emulator.

Other than that, designing the interface of the application was quite difficult as the design must be responsive to the size of the device. If not done properly, the design of the overall application can become a mess as some of the elements move with different screen sizes. To combat this, creating a guideline to make sure the elements stay in place was the key to ensure a responsive user interface. Overall, the experience when developing was very fun and interesting as new things were discovered and was able to become new knowledge.

4.6 Summary

To summarise, this chapter covers the implementation of the mobile application as well as the functionality testing which shows what are the results of the developed mobile application. The interfaces were also shown to give a more insight into what the application looks like and see some functions that was implemented. The discussions were also done to give insight into the process of developing the application.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

5.1 Revisiting Objectives

Revisiting the objectives is crucial in comparing the proposed objectives with the actual development of the mobile application. This section will also cover how the implementation of the traceability system is done to track where the palm fruit oil came from using a QR code generated after harvesting by farmers. The objectives are:

- a. To develop a suitable machine learning algorithm to detect the ripeness of the palm fruit.
- b. To implement a traceability system to track the process of a palm fruit.
- c. To test the functionality of palm fruit ripeness detection with traceability system.

5.1.1 Development of Machine Learning Algorithm to Detect the Ripeness of Palm Fruit

The project is aimed to develop a mobile application that can detect the different stages of ripeness in palm fruit oil like under ripe, unripe, ripe, and over ripe. To develop such an application, determining a suitable ML algorithm is key in reaching this goal. In the literature review, a comparison between different ML algorithms was conducted between SVM, KNN and CNN. From the research, CNN was chosen as the best algorithm out of the rest due to its excellent performance in classifying objects in images and has the highest accuracy rate between SVM and KNN.

To develop the CNN algorithm, finding the dataset is crucial to ensure that when training the model, we have the necessary data that the model can train to. The dataset used to create the project is an online dataset which has multiple images of palm fruit oil in different stages. The ripeness stage that was chosen for this project are unripe, under ripe, ripe and over ripe. Once the dataset is found, creating the model is done using an IDE such as jupyter lab which helps during the development of the CNN algorithm because of its ability to compile and run different lines of codes in one document. Tensorflow and Keras are used to create the ML features of this project

which is done in Python programming language.

The CNN algorithm was successfully developed with an accuracy of 95% and up to 250 epochs to train the model. After the model was saved, it was converted to a TFLite model to allow the model to work in mobile applications.

5.1.2 Implementation of traceability system function on the transportation of Palm Fruit

The second objective of this project is to implement a traceability system that can track where the palm fruit bunches originated from. After farmers detect the ripeness for each batch, they will key in the information of the batch like the ripeness, the location of the palm fruit harvested and date it was harvested. After the information is put into the database a QR code will be generated based on the id of the batch that was just inserted into the database.

Then the QR code will be updated to the table as well. Once the batches are sent to the mill, the staff there will then get the data of each batch by scanning the QR code. If the batch's ripeness is under ripe, unripe or over ripe, the batch will be graded as suitable for animal feed or fertilizer while ripe palm fruit oil will be weight and the average weight of the batch will be inserted into the database. The ripe palm fruit oil will then be graded as good for oil production and sent to the mill's oil production section. Next, the staff will weigh the palm fruit oil and key in the average weight of the batch.

5.1.3 Testing the Functionality of the Mobile Application

The final objective of the project is to test the functionalities of the mobile application to ensure it meets the desired performance criteria. This application incorporates a variety of features, including user login, ripeness detection of palm fruits, and grading the palm fruit oil. The development process involved using Android Studio and Java to build the application framework while for the implementation of ripeness detection feature, the TFLite model is employed. This enables the application to detect the ripeness of the palm fruit oil using images from the gallery or even using the camera to capture palm fruit oil image in real time. The traceability system was also tested where the QR code is generated after filling in the batch information based on the id for each

batch.

5.2 Strengths and Limitations

5.2.1 Strengths

The mobile application that was developed has several strengths that enhance its utility and effectiveness. Firstly, the application uses a QR-based traceability system, allowing for precise tracking of palm fruits from the farm to the mill. The application uses machine learning to identify the ripeness of palm fruits with that has a 95% accuracy rate. Additionally, it assesses the quality of the oil produced based on the weight and ripeness of the fruits. The application also provides guidance on the potential uses for unripe, under-ripe, and overripe fruits, making sure the staff know what the different uses of all the palm fruit ripeness stages.

5.2.2 Limitations

The main drawback of the developed application is that the database is not updated in real-time, which can cause lost data when a user deletes the application from their phone. This limitation might affect the overall efficiency and reliability of the system, particularly in scenarios where real time data is crucial.

Other than that, the dataset used is also a limitation due to it being a dataset that was found online. This is because, when requesting the Malaysian Palm Oil Board (MPOB) for their dataset of palm fruit in different ripeness stages, they do not allow the dataset to be given due to it being confidential. Although, the dataset is from an online source, the images do represent the palm fruits that are generally found in Malaysia making the project to be more in line.

5.3 Future Work Recommendations

Based on the issue that was mentioned in the above section, to improve the application's ability to keep and update the data in real time, using a different database is needed. One of the suggested databases is to use Firebase Realtime Database which helps to keep the data updated quickly and accurately. Another suggestion is to implement MySQL to the mobile application because of its similarity to SQLite.

5.4 Summary

In conclusion, revisiting the project objectives highlights the development and implementation of a mobile application aimed at detecting the ripeness of palm fruits and providing traceability from farm to mill. The project involved developing a CNN algorithm with a 95% accuracy rate for ripeness detection, implemented using TensorFlow, Keras, and TFLite for mobile compatibility. The application also features a QR-based traceability system to track palm fruit batches. Developed using Android Studio and Java, the app includes user login, ripeness detection, and quality grading of oil produced. While the app excels in machine learning accuracy and traceability, its primary limitation is the lack of real-time database updates, potentially leading to data loss. To address this, transitioning to databases like Firebase Realtime Database or MySQL is recommended to ensure real-time data accuracy and reliability.

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APPENDICES

APPENDIX 1

Gantt Chart for Semester 5

Week															
Activities	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Contact Potential Supervisors															
F1 (Mutual Acceptance Form)															
Chapter 1 Presentation															
F2 (Project Motivation Evaluation Form)+ Presentation slides															
Chapter 1															
Chapter 2 + F3 (Literature Review Evaluation Form)															
Chapter 3 + F4 (Methodology Evaluation Form)															
Submission of Full Report															
Final Proposal Presentation															

Gantt Chart for Semester 6

Week															
Activities	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Submission of Chapter 1 (Refined)															
Submission of Chapter 2 (Refined)															
Submission of Chapter 3 (Refined)															
Submission of Business Model Canvas															
Submission of Chapter 4															
Progress Presentation															
Submission of Chapter 5															
Submission of First Draft															
Submission of Full Report															
Presentation of Final Year Project															
Plagiarism Checking															
Development of Mobile Application															